American International University-Bangladesh

Faculty of Science & Technology (FST)

Department of Computer Science

Introduction to Data Science

Mid-Term Project Report

Fall 2025-2026

Section: I, Group: G

Data Analysis and Preprocessing of Netflix Movies and TV Shows Dataset

Supervised by: Kamrun Naher Koli

SL #	Student Name	Student ID	Contribution (%)
1.	Dip Khastagir	23-50346-1	33.4%
2.	Shuvra Kanti Das	22-49496-3	33.3%
3	Chandrima Dam	22-49509-3	33.3%

Table of Contents

1. Title Page
2. Table of Contents
3. Introduction
4. Dataset Description
5. Data Issues Identified
6. Data Cleaning & Preparation
    - 6.1 Installing Required Packages
    - 6.2 Loading Libraries
    - 6.3 Loading the Dataset
    - 6.4 Missing Values Detection
    - 6.5 Converting Empty Strings to NA
    - 6.6 Handling Missing Categorical Values
    - 6.7 Missing Value Visualization
    - 6.8 Duplicate Detection & Removal
    - 6.9 Invalid Duration Detection
    - 6.10 Parsing Release Date
    - 6.11 Extracting Release Year
    - 6.12 Outlier Detection & Removal
    - 6.13 Converting Categorical to Numeric
    - 6.14 Converting Numeric to Categorical
    - 6.15 Normalization
    - 6.16 Data Filtering
7. Balancing the Dataset
8. Train–Test Split
9. Exploratory Data Analysis
10. Descriptive Statistics
11. Conclusion
12. References

Introduction

In the era of digital streaming, platforms like Netflix generate vast amounts of data on user preferences, content production, and global reach. This project aims to analyze a comprehensive Netflix dataset containing information about movies and TV shows, including attributes such as Category, Director, Cast, Country, Rating, Duration, and Release Date. The primary goal is to clean, preprocess, and explore this dataset to uncover trends and patterns in Netflix's content library.

The workflow begins with handling missing or inconsistent data, removing duplicates, detecting invalid entries, and parsing dates to extract meaningful features like release year. Outliers are identified and removed to ensure the reliability of analysis. Categorical variables are transformed into numeric and grouped forms to facilitate modeling and statistical analysis, while numerical features are normalized for consistency.

Further, the dataset is balanced using upsampling to address class imbalances between Movies and TV Shows, and it is split into training and testing sets to support future predictive modeling. Comprehensive exploratory data analysis, including descriptive statistics and visualizations such as bar plots, histograms, and boxplots, is performed to understand distributions, trends, and relationships in the dataset.

Ultimately, this project produces a clean, well-prepared dataset that not only provides insights into Netflix's content patterns over time and across categories but also lays the foundation for further predictive modeling and data-driven decision-making.

Dataset Description

The Netflix dataset contains information about movies and TV shows available on Netflix. Each entry includes details such as title, type (Movie or TV Show), director, cast, country, date added, release year, rating, duration, genres, and a brief description. The dataset has over 8,800 titles and includes both numeric and categorical data, making it suitable for analysis of trends over time, content distribution by type, genre, country, and ratings. Some entries have missing values, so data cleaning is necessary before analysis. This dataset is ideal for exploring patterns in Netflix content, visualizing release trends, and performing classification or clustering tasks.

Data Issues Identified

During the initial exploration of the Netflix dataset, several data quality problems were identified that required correction before analysis.

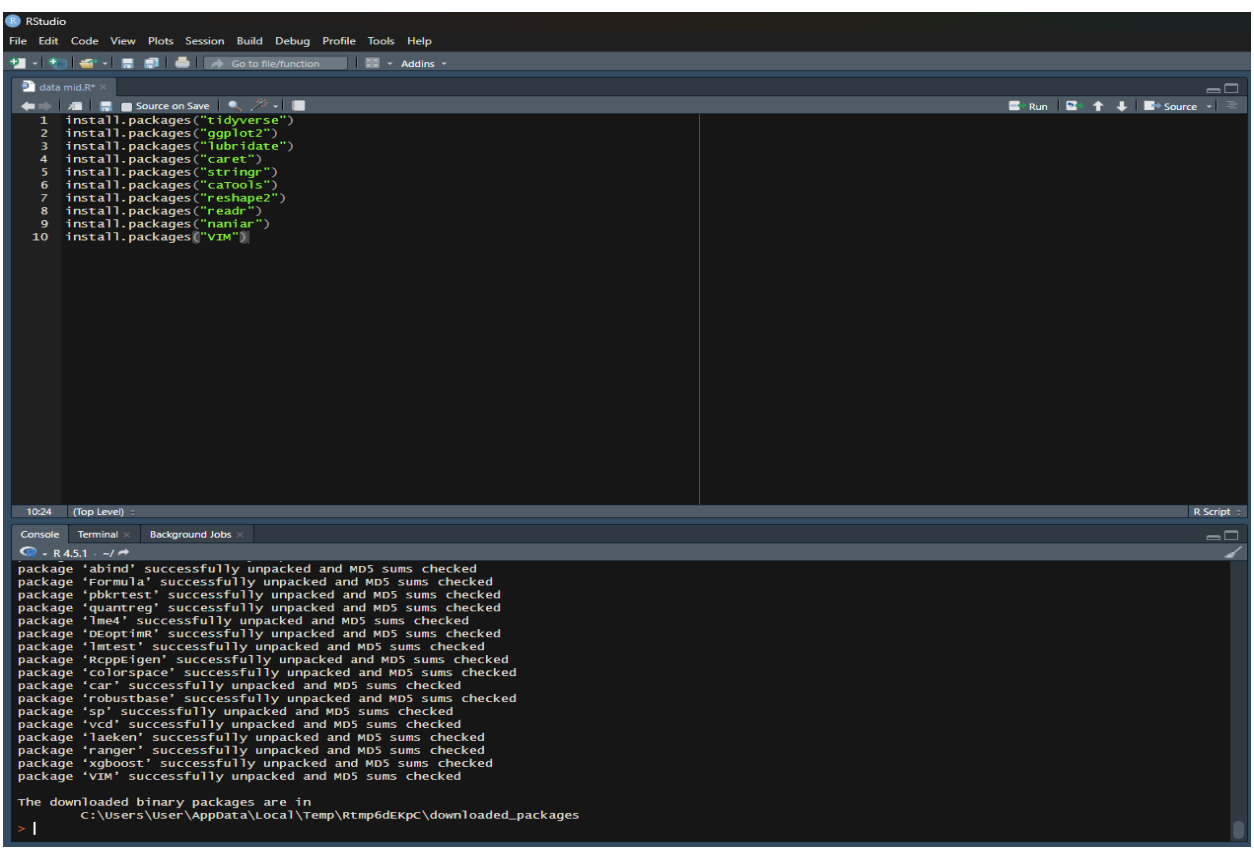
- **Missing Values:** Important fields such as Director, Cast, Country, and Rating contained missing entries, often recorded as empty strings instead of NA, making them harder to detect and handle.
- **Inconsistent Release Date Formats:** The Release_Date column used multiple date formats such as 14-Aug-20, January 5, 2019, and 2020-03-15, preventing direct conversion into a standard date format.
- **Duplicate Records:** Some titles appeared more than once in the dataset, which could distort summary statistics and needed to be removed for accurate analysis.
- **Invalid Duration Values:** A few Duration entries did not match expected patterns like "90 min" for movies or "2 Seasons" for TV shows, indicating irregular or unusable values.
- **Outliers in Release Year:** After extracting the Release Year, some values fell outside reasonable ranges, likely due to data entry errors, and were removed to avoid skewed analysis.
- **Mixed Data Types:** Several variables required conversion, such as converting the Movie/TV Show label into a numeric target and grouping the release year into categorical ranges.
- **Imbalanced Target Classes:** Movies significantly outnumbered TV shows, creating class imbalance; this required balancing through upsampling to ensure fair representation of both categories.

Data Cleaning & Preparation

1. **Installing Required Packages**

```
install.packages("tidyverse")
install.packages("ggplot2")
install.packages("lubridate")
install.packages("caret")
install.packages("stringr")
install.packages("caTools")
install.packages("reshape2")
install.packages("readr")
install.packages("naniar")
install.packages("VIM")
```

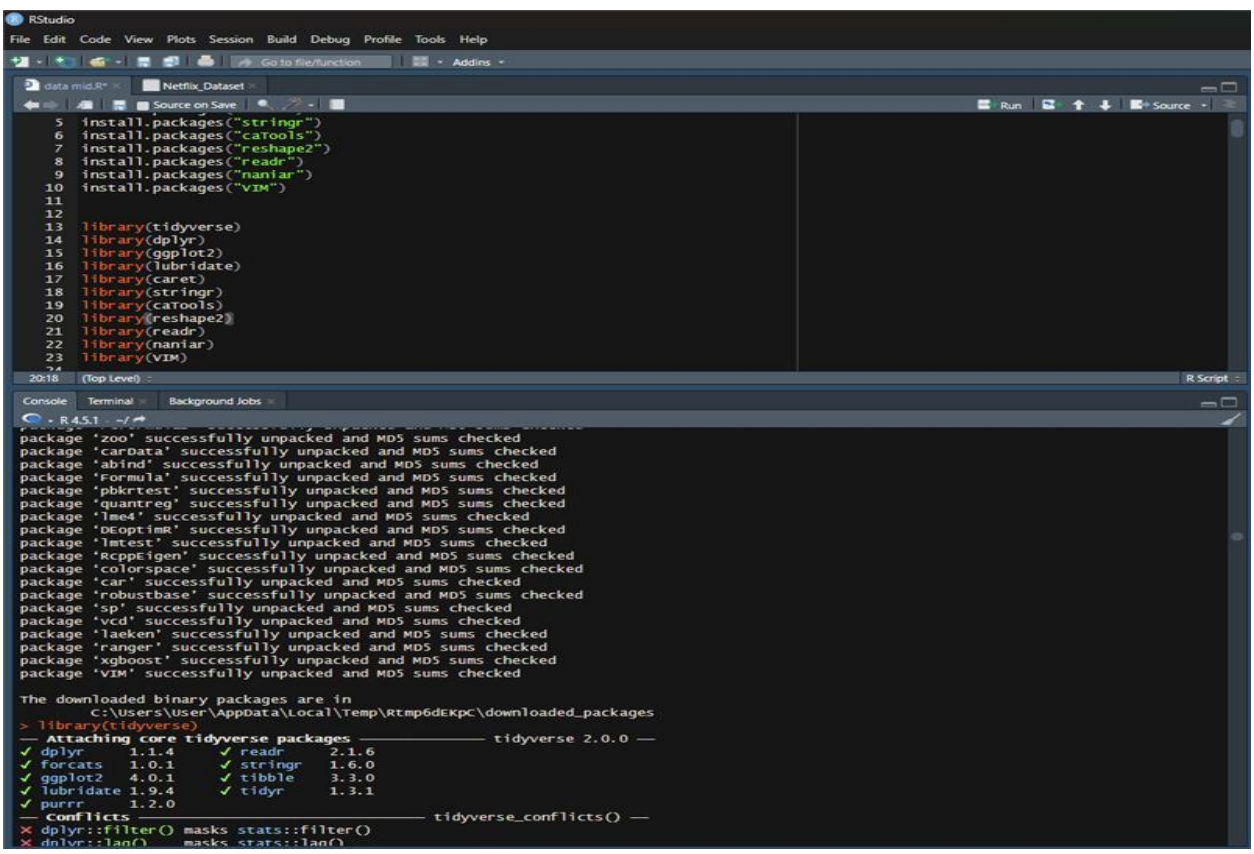
The project begins by installing and loading several R libraries. These libraries are like special toolboxes that give R extra powers for tasks such as data cleaning, visualization, date handling, and building models. Packages like tidyverse, dplyr, and ggplot2 help with data manipulation and plotting, while lubridate handles dates, and others assist with reshaping data, checking missing values, and splitting data for modeling.



2. **Loading Libraries**

```
library(tidyverse)
library(dplyr)
library(ggplot2)
library(lubridate)
library(caret)
library(stringr)
library(caTools)
library(reshape2)
library(readr)
library(naniar)
library(VIM)
```

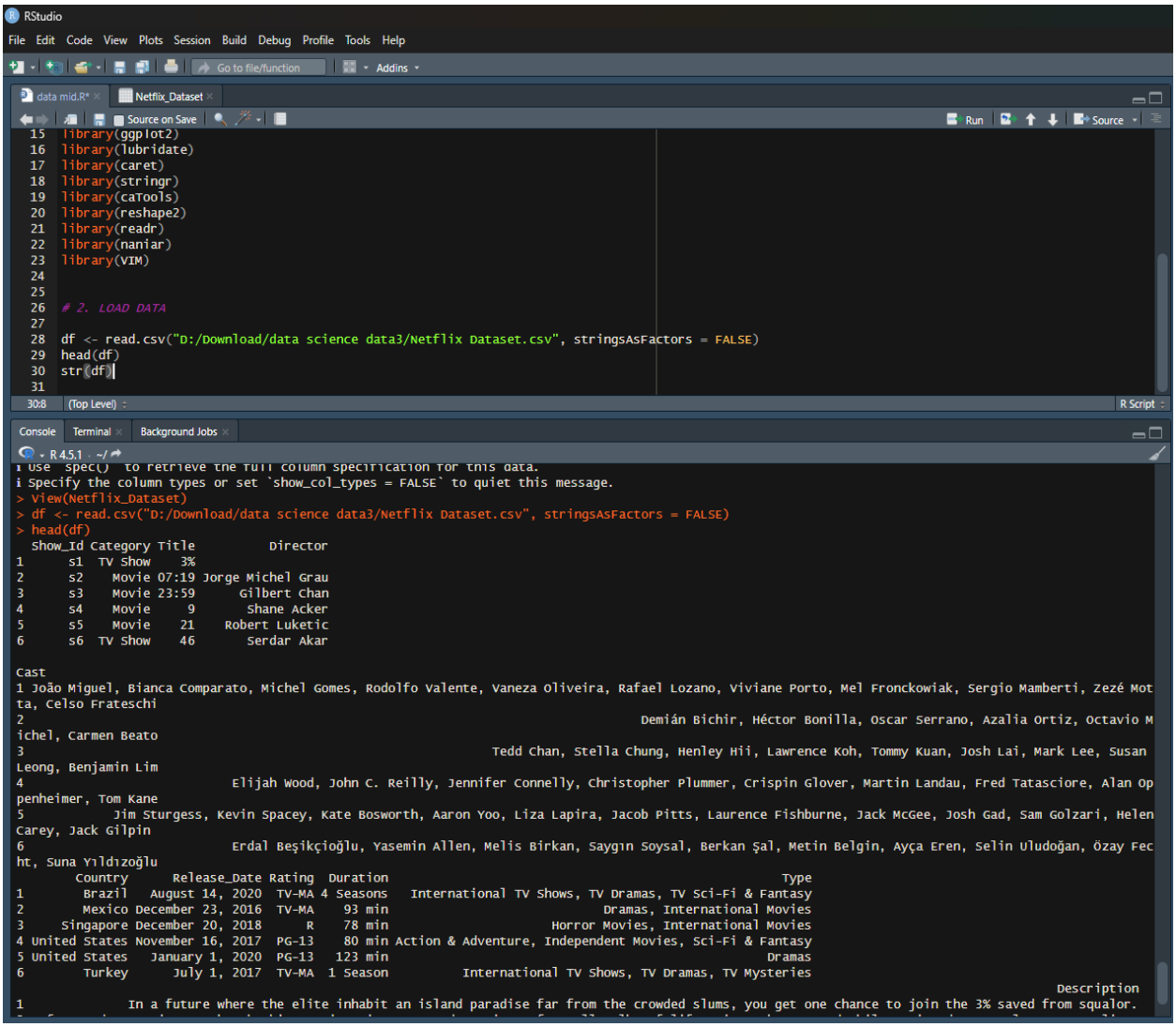
This section installs and loads all necessary R packages required for data cleaning, manipulation, visualization, and analysis. Packages such as tidyverse, dplyr, and ggplot2 help with data wrangling and graphing, while lubridate is used for date handling. naniar and VIM support missing-data visualization, and caTools is used later for splitting the dataset. Loading these libraries ensures all required functions are available throughout the analysis.



3. **Loading the Dataset**

```
df <- read.csv("D:/IDS/Netflix Dataset.csv", stringsAsFactors = FALSE)
head(df)
str(df)
```

The dataset is imported into R from the specified file path using read.csv. After loading, the first few rows (head(df)) and the structure of the dataset (str(df)) are displayed to understand the column types, data formats, and general layout. This initial inspection helps identify potential inconsistencies or cleaning needs.



RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins

data mid.R* | Netflix_Dataset

Source on Save | Run | Source

```r
library(ggplot2)
library(lubridate)
library(caret)
library(stringr)
library(caTools)
library(reshape2)
library(readr)
library(naniar)
library(VIM)


# 2. LOAD DATA

df <- read.csv("D:/Download/data science data3/Netflix Dataset.csv", stringsAsFactors = FALSE)
head(df)
str(df)

```

30:1 (Top Level) R Script

Console | Terminal | Background Jobs

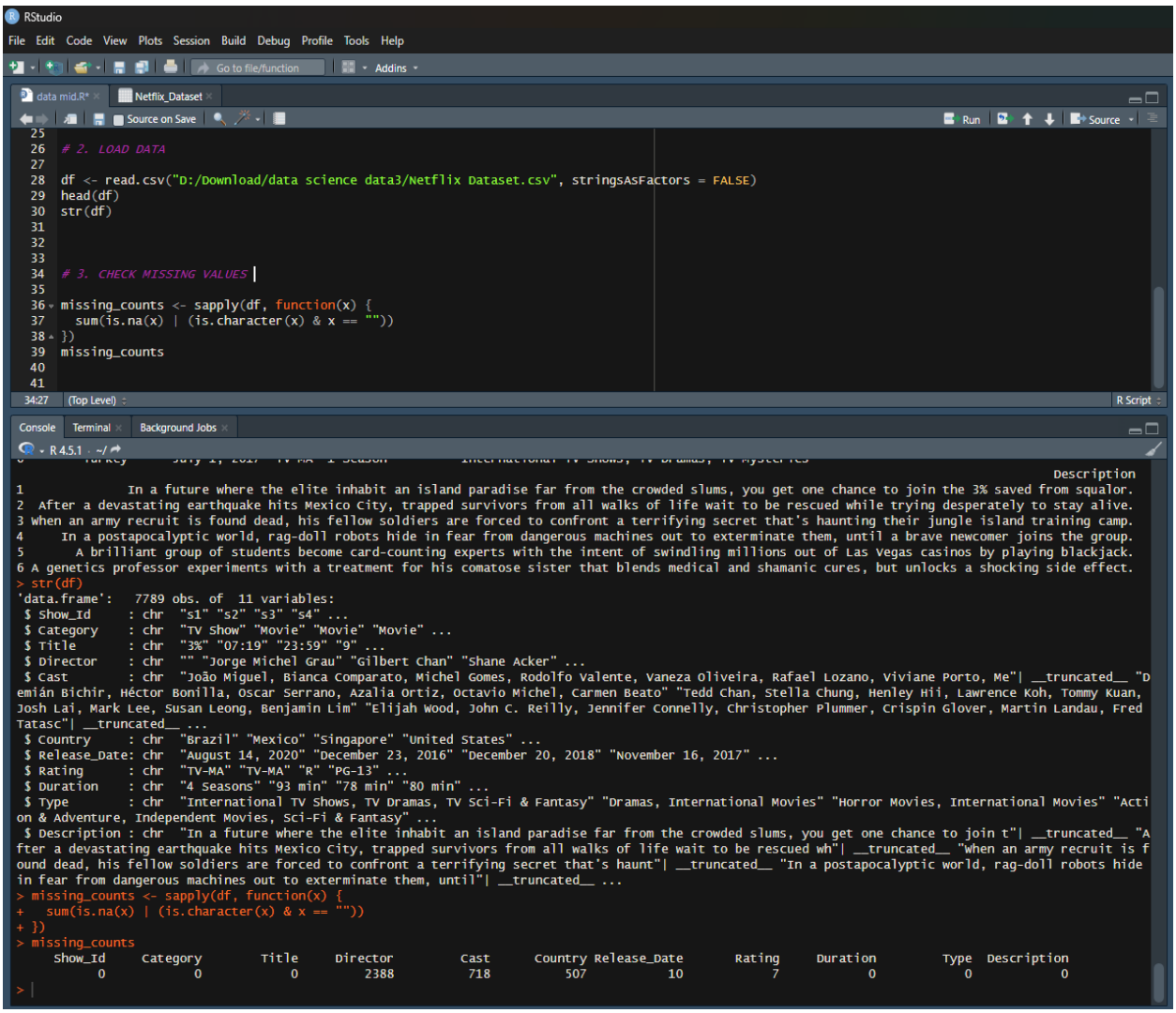
R 4.5.1 · ~/

```
        Country  Release_Date Rating  Duration                                                  Type
1        Brazil   August 14, 2020  TV-MA 4 Seasons   International TV Shows, TV Dramas, TV Sci-Fi & Fantasy
2        Mexico December 23, 2016  TV-MA    93 min                                Dramas, International Movies
3     Singapore December 20, 2018      R    78 min                           Horror Movies, International Movies
4 United States November 16, 2017  PG-13    80 min Action & Adventure, Independent Movies, Sci-Fi & Fantasy
5 United States   January 1, 2020  PG-13   123 min                                                     Dramas
6        Turkey      July 1, 2017  TV-MA  1 Season          International TV Shows, TV Dramas, TV Mysteries
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
1        Description
2 In a future where the elite inhabit an island paradise far from the crowded slums, you get one chance to join the 3% saved from squalor.
3 After a devastating earthquake hits Mexico City, trapped survivors from all walks of life wait to be rescued while trying desperately to stay alive.
4 When an army recruit is found dead, his fellow soldiers are forced to confront a terrifying secret that's haunting their jungle island training camp.
5 In a postapocalyptic world, rag-doll robots hide in fear from dangerous machines out to exterminate them, until a brave newcomer joins the group.
6 A brilliant group of students become card-counting experts with the intent of swindling millions out of Las Vegas casinos by playing blackjack.
6 A genetics professor experiments with a treatment for his comatose sister that blends medical and shamanic cures, but unlocks a shocking side effect.
> str(df)
'data.frame':   7789 obs. of  11 variables:
 $ Show_Id     : chr  "s1" "s2" "s3" "s4" ...
 $ Category    : chr  "TV Show" "Movie" "Movie" "Movie" ...
 $ Title       : chr  "3%" "07:19" "23:59" "9" ...
 $ Director    : chr  "" "Jorge Michel Grau" "Gilbert Chan" "Shane Acker" ...
 $ Cast        : chr  "João Miguel, Bianca Comparato, Michel Gomes, Rodolfo Valente, Vaneza Oliveira, Rafael Lozano, Viviane Porto, Me"| __truncated__ "D
emián Bichir, Héctor Bonilla, Oscar Serrano, Azalia Ortiz, Octavio Michel, Carmen Beato" "Tedd Chan, Stella Chung, Henley Hii, Lawrence Koh, Tommy Kuan,
Josh Lai, Mark Lee, Susan Leong, Benjamin Lim" "Elijah Wood, John C. Reilly, Jennifer Connelly, Christopher Plummer, Crispin Glover, Martin Landau, Fred
Tatasc"| __truncated__ ...
 $ Country     : chr  "Brazil" "Mexico" "Singapore" "United States" ...
 $ Release_Date: chr  "August 14, 2020" "December 23, 2016" "December 20, 2018" "November 16, 2017" ...
 $ Rating      : chr  "TV-MA" "TV-MA" "R" "PG-13" ...
 $ Duration    : chr  "4 Seasons" "93 min" "78 min" "80 min" ...
 $ Type        : chr  "International TV Shows, TV Dramas, TV Sci-Fi & Fantasy" "Dramas, International Movies" "Horror Movies, International Movies" "Acti
on & Adventure, Independent Movies, Sci-Fi & Fantasy" ...
 $ Description : chr  "In a future where the elite inhabit an island paradise far from the crowded slums, you get one chance to join t"| __truncated__ "A
fter a devastating earthquake hits Mexico City, trapped survivors from all walks of life wait to be rescued wh"| __truncated__ "When an army recruit is f
ound dead, his fellow soldiers are forced to confront a terrifying secret that's haunt"| __truncated__ "In a postapocalyptic world, rag-doll robots hide
in fear from dangerous machines out to exterminate them, until"| __truncated__ ...
>
```

4. **Missing Values Detection**

```
missing_counts <- sapply(df, function(x) {
  sum(is.na(x) | (is.character(x) & x == ""))
})
missing_counts
```

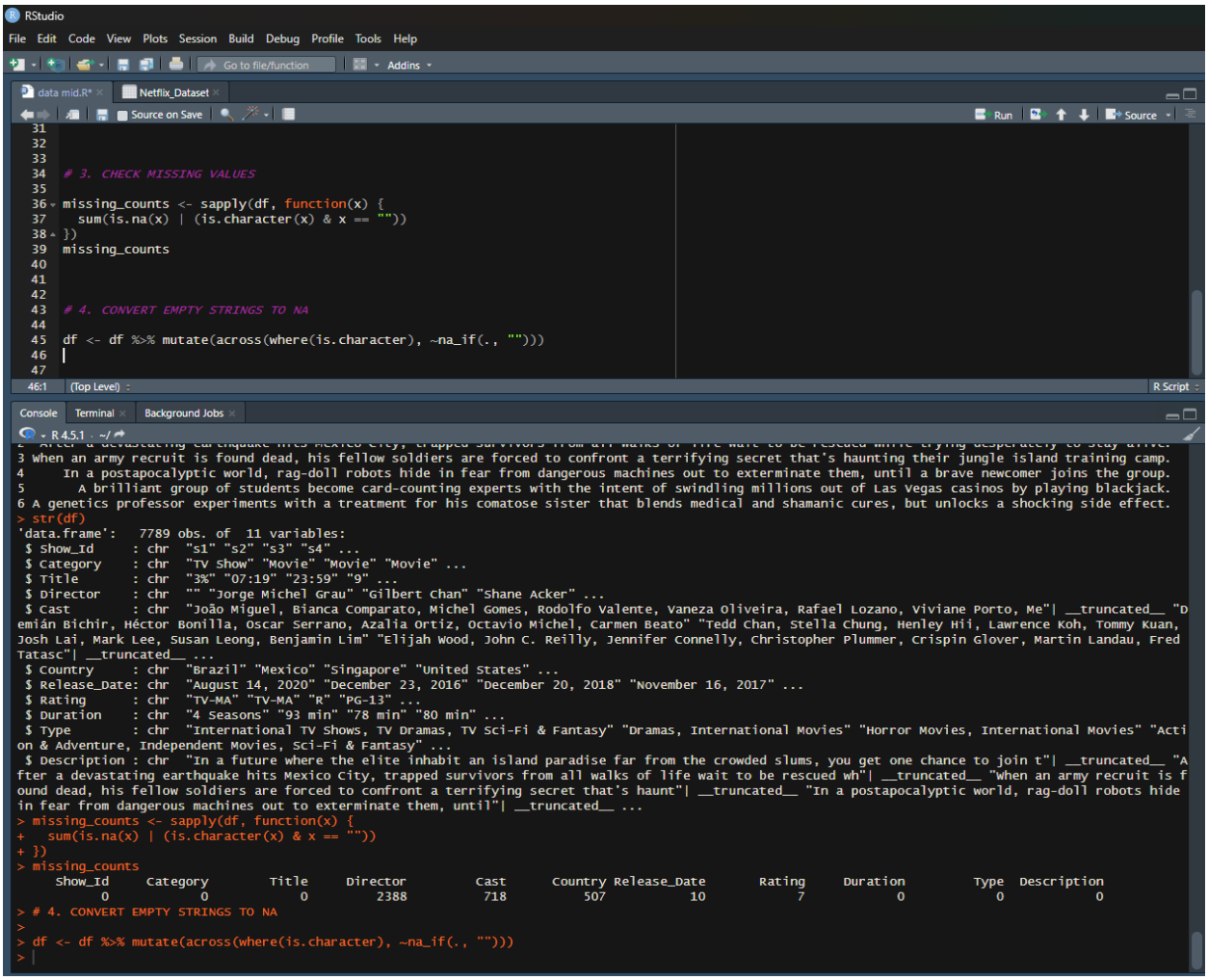
Since missing values can appear as either NA or empty strings, this step calculates the total number of missing entries in each column using a safe approach that checks both conditions. This provides a clear understanding of which variables contain missing information and require further cleaning.



5. **Convert Empty Strings to NA**

   df <- df %>% mutate(across(where(is.character), ~na_if(., "")))

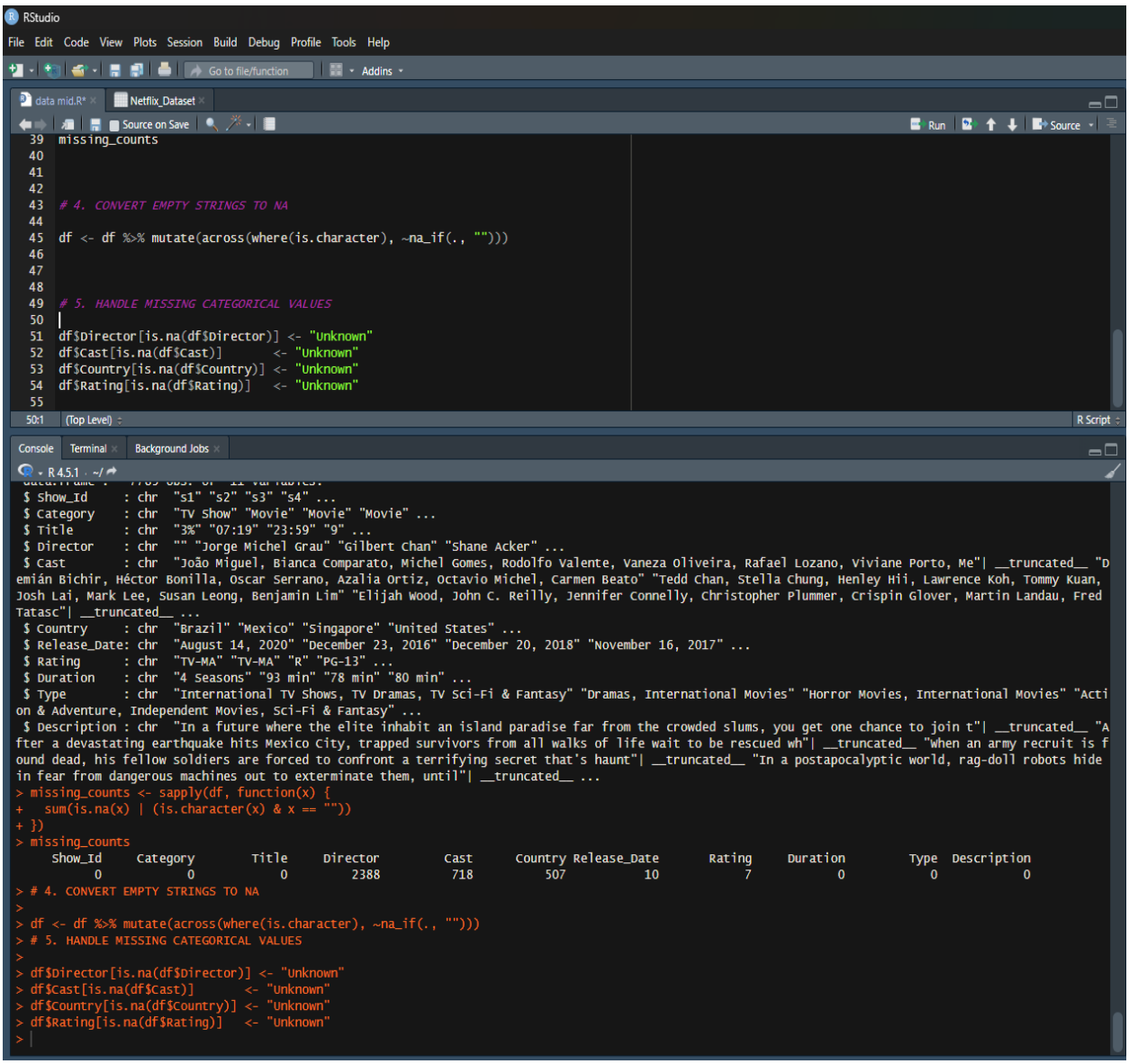
Empty strings (“”) in character columns are replaced with proper NA values. This ensures all missing information is uniformly represented, making it easier to handle, replace, or visualize missing data during subsequent cleaning steps.



6. **Categorical Missing Value Handling**

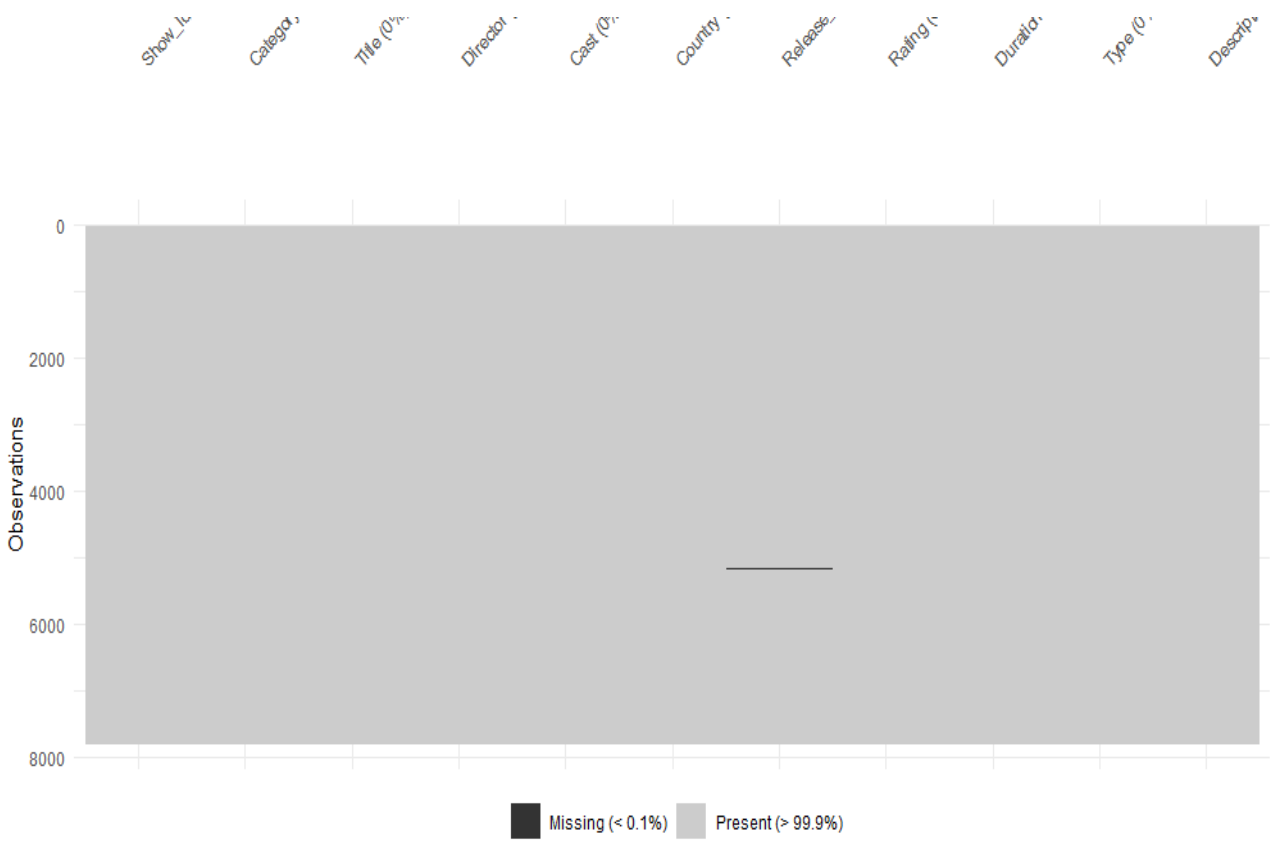
```
df$Director[is.na(df$Director)] <- "Unknown"
df$Cast[is.na(df$Cast)]       <- "Unknown"
df$Country[is.na(df$Country)] <- "Unknown"
df$Rating[is.na(df$Rating)]   <- "Unknown"
```

Missing values in the categorical fields Director, Cast, Country, and Rating are replaced with the placeholder value "Unknown." This prevents issues in later stages such as grouping, filtering, or plotting, where missing categories could cause errors or distort analysis.



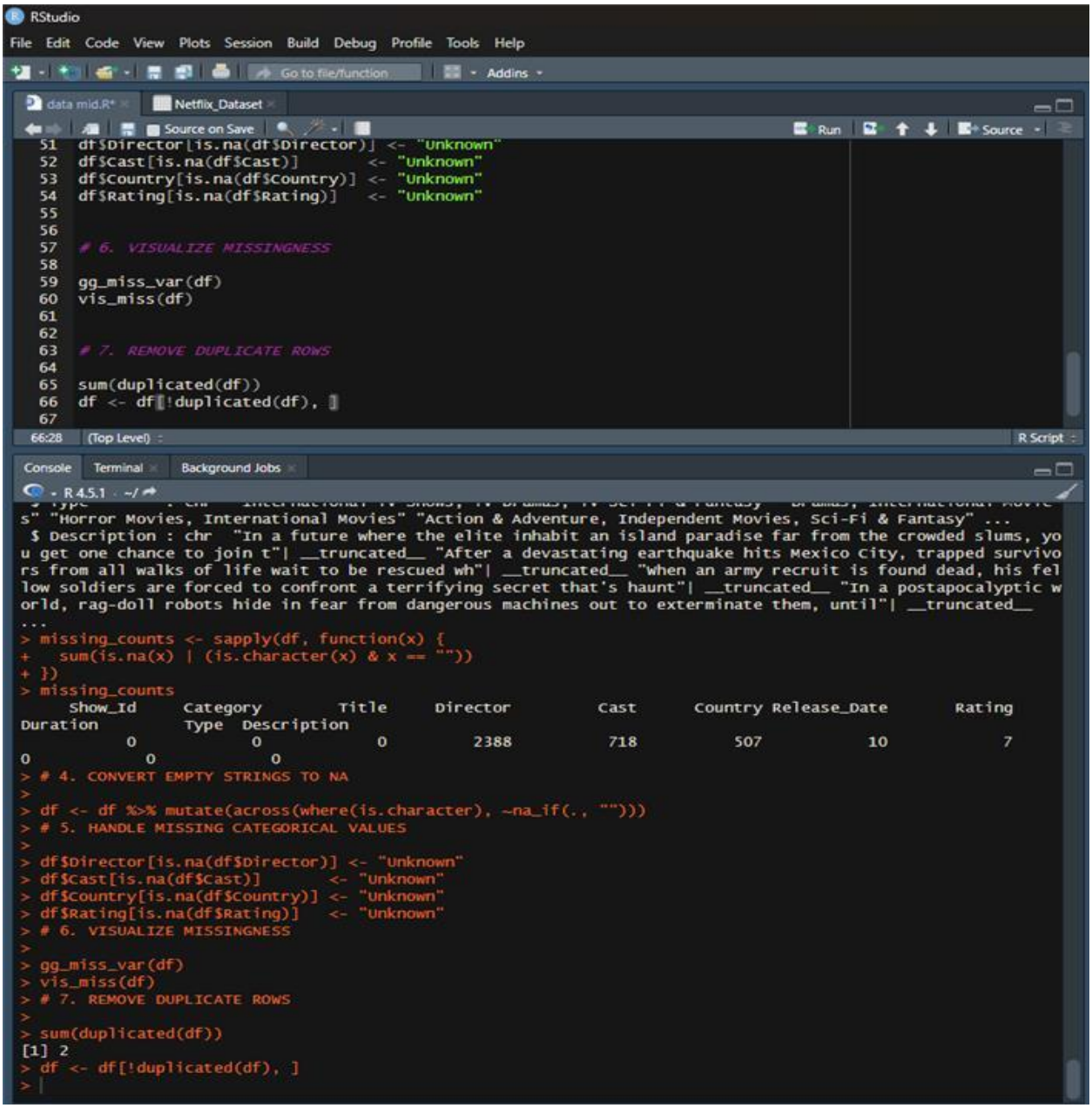
7. **Missing Value Visualization**
   gg_miss_var(df)
   vis_miss(df)

Missing-data patterns are visualized using gg_miss_var() and vis_miss(). These plots help identify which columns contain missing values and whether the missingness follows any pattern. Visual inspection provides a clearer understanding of data quality issues than numerical counts alone.



8. **Duplicate Detection & Removal**
   sum(duplicated(df))
   df <- df[!duplicated(df),]

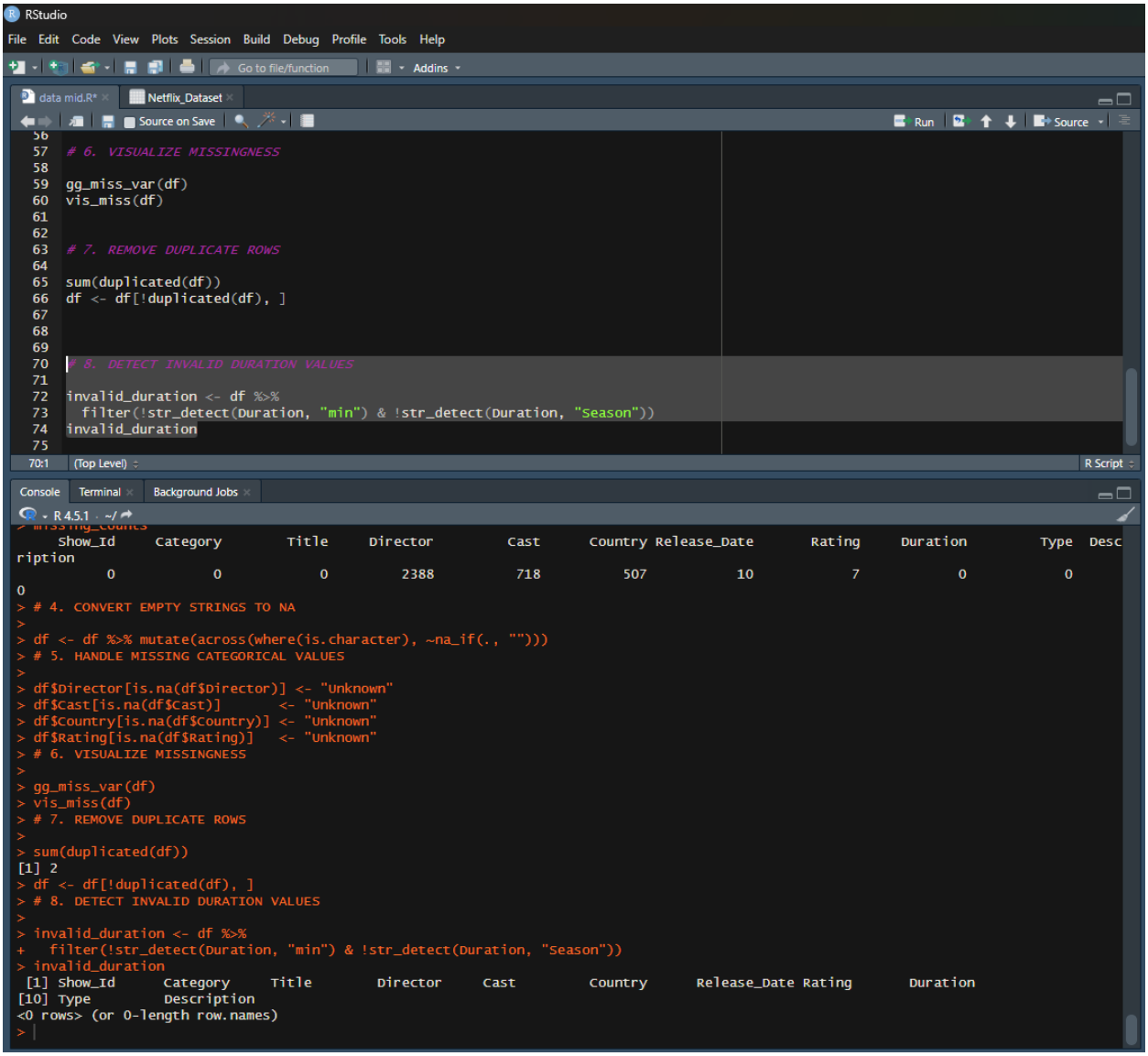
Duplicate entries can bias analysis by artificially inflating counts or skewing distributions. This block identifies duplicate rows using duplicated(df) and removes them to ensure all remaining data points represent unique records.



9. **Invalid Duration Detection**

```
invalid_duration <- df %>%
  filter(!str_detect(Duration, "min") & !str_detect(Duration, "Season"))
invalid_duration
```

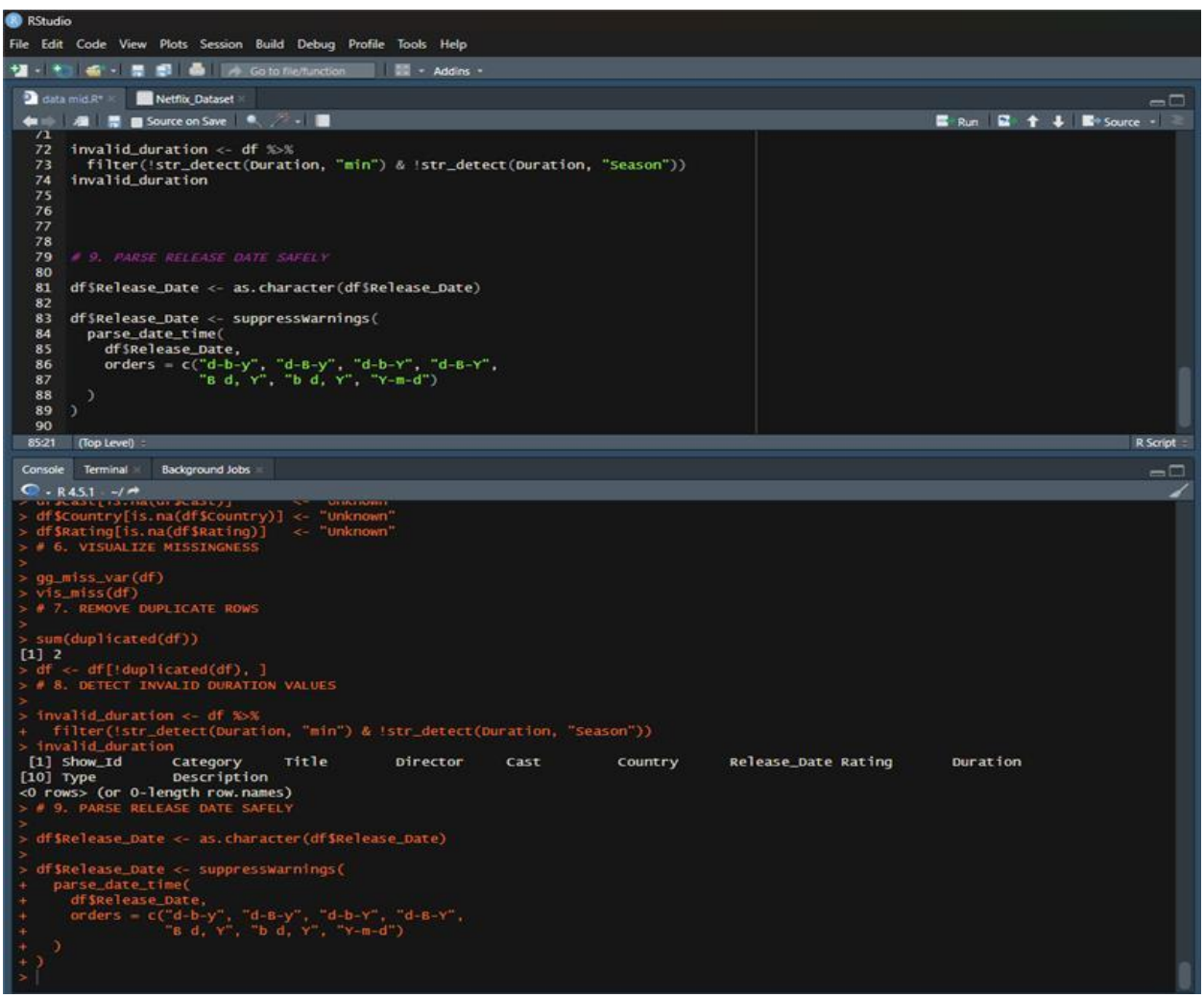
The Duration column should contain values ending in “min” for movies or “Season/Seasons” for TV shows. Any record that does not match these formats is flagged as invalid. Detecting such anomalies helps ensure the Duration field is consistent and meaningful.



10. Release Date Parsing

```
df$Release_Date <- as.character(df$Release_Date)

df$Release_Date <- suppressWarnings(
  parse_date_time(
    df$Release_Date,
    orders = c("d-b-y", "d-B-y", "d-b-Y", "d-B-Y",
               "B d, Y", "b d, Y", "Y-m-d")
  )
)
```

The Release_Date column contains values in multiple different formats. This step standardizes all dates using parse_date_time() while suppressing warnings. Converting these strings into actual date objects enables reliable extraction of the release year and supports time-based analysis.

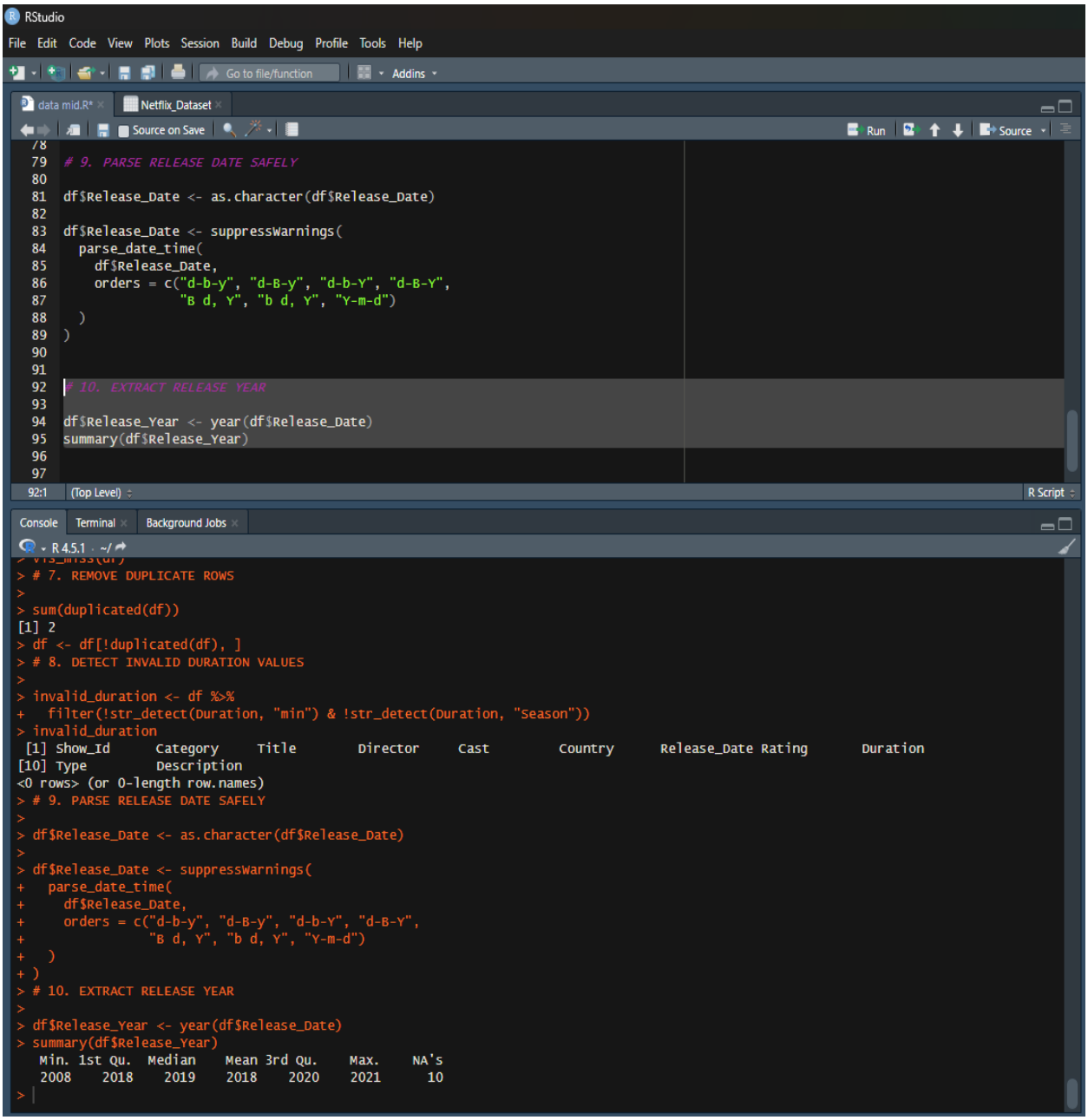


11. Extracting Release Year

df$Release_Year <- year(df$Release_Date)

summary(df$Release_Year)

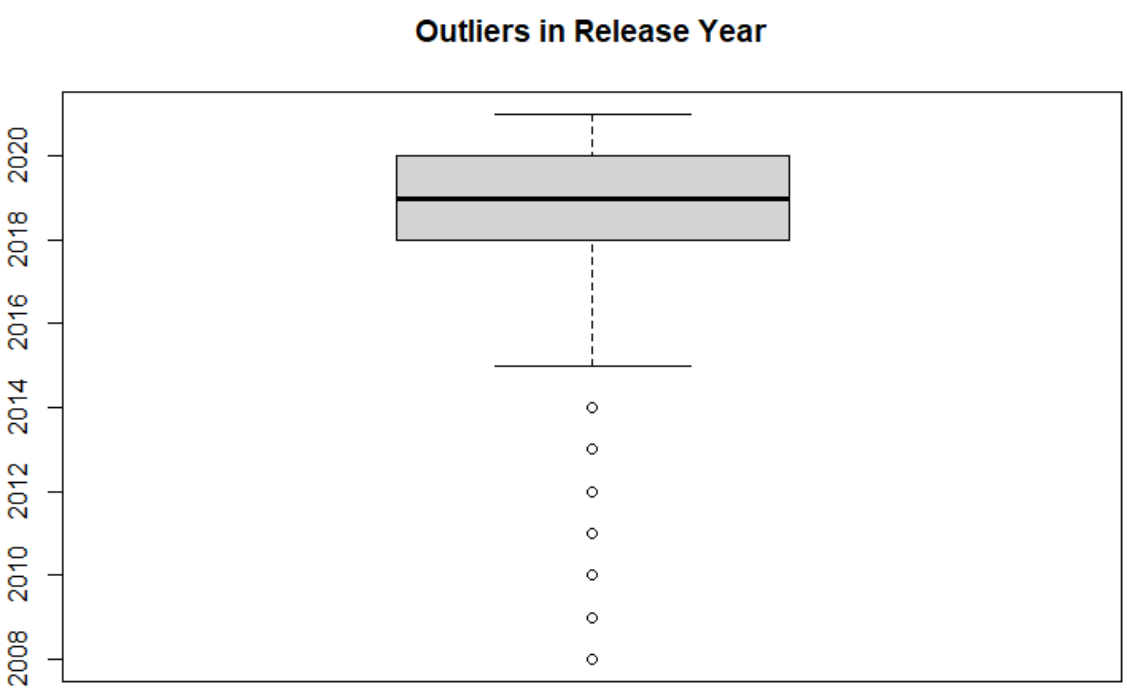
Once the Release_Date column is converted into a proper date format, the year() function extracts the year component into a new variable, Release_Year. This numeric field is used later for outlier detection, grouping, and visualization.



12. Outlier Detection & Removal

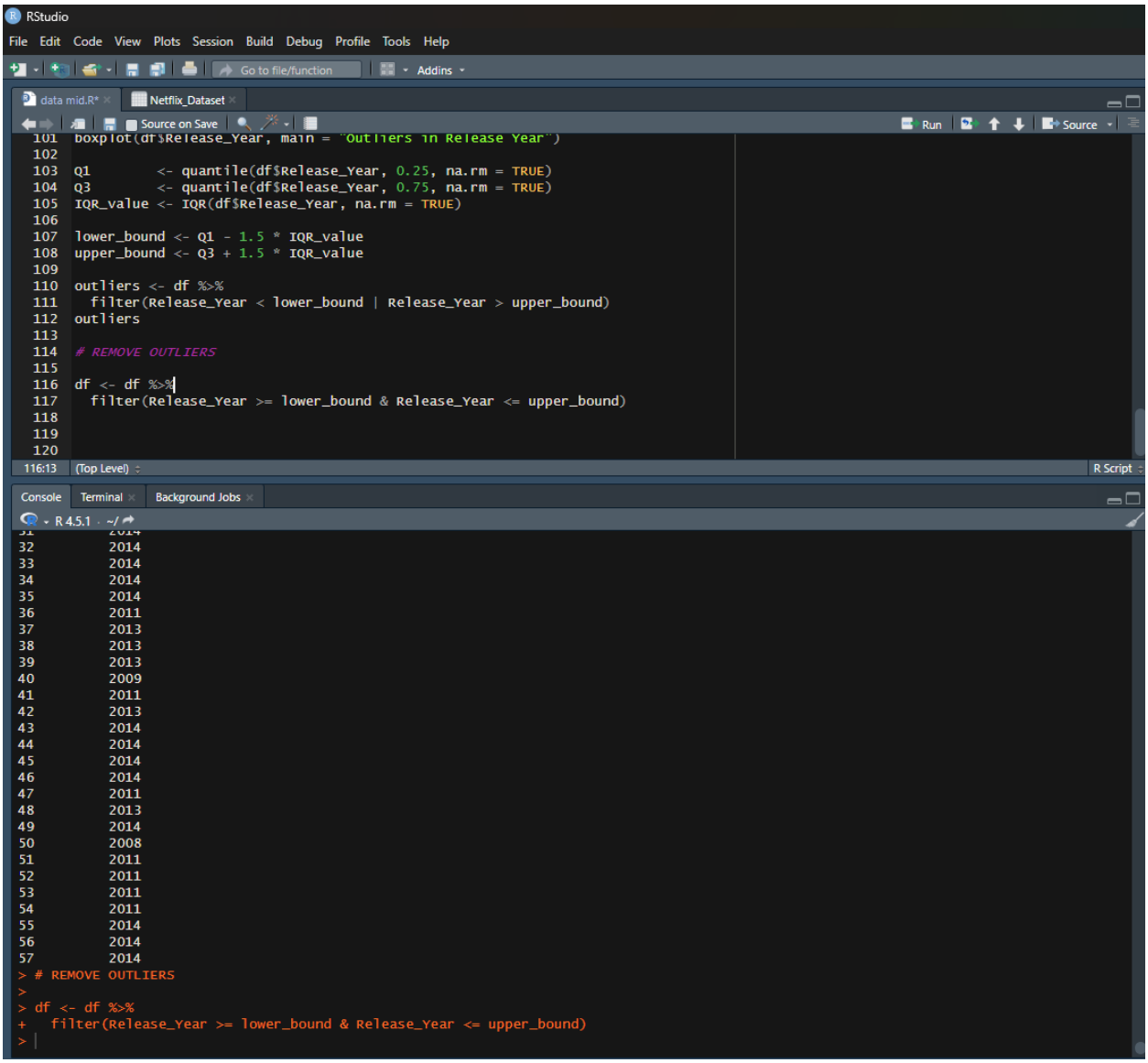
```
# DETECT OUTLIERS IN RELEASE YEAR
boxplot(df$Release_Year, main = "Outliers in Release Year")

Q1 <- quantile(df$Release_Year, 0.25, na.rm = TRUE)
Q3 <- quantile(df$Release_Year, 0.75, na.rm = TRUE)
IQR_value <- IQR(df$Release_Year, na.rm = TRUE)

lower_bound <- Q1 - 1.5 * IQR_value
upper_bound <- Q3 + 1.5 * IQR_value

outliers <- df %>%
  filter(Release_Year < lower_bound | Release_Year > upper_bound)
outliers
```

Outliers are detected using the Interquartile Range (IQR) method. A boxplot visualizes unusual values, and lower and upper bounds are calculated. Any release years falling outside this acceptable range are considered outliers and removed to prevent skewed statistical results.



```
# REMOVE OUTLIERS
df <- df %>%
  filter(Release_Year >= lower_bound & Release_Year <= upper_bound)
```

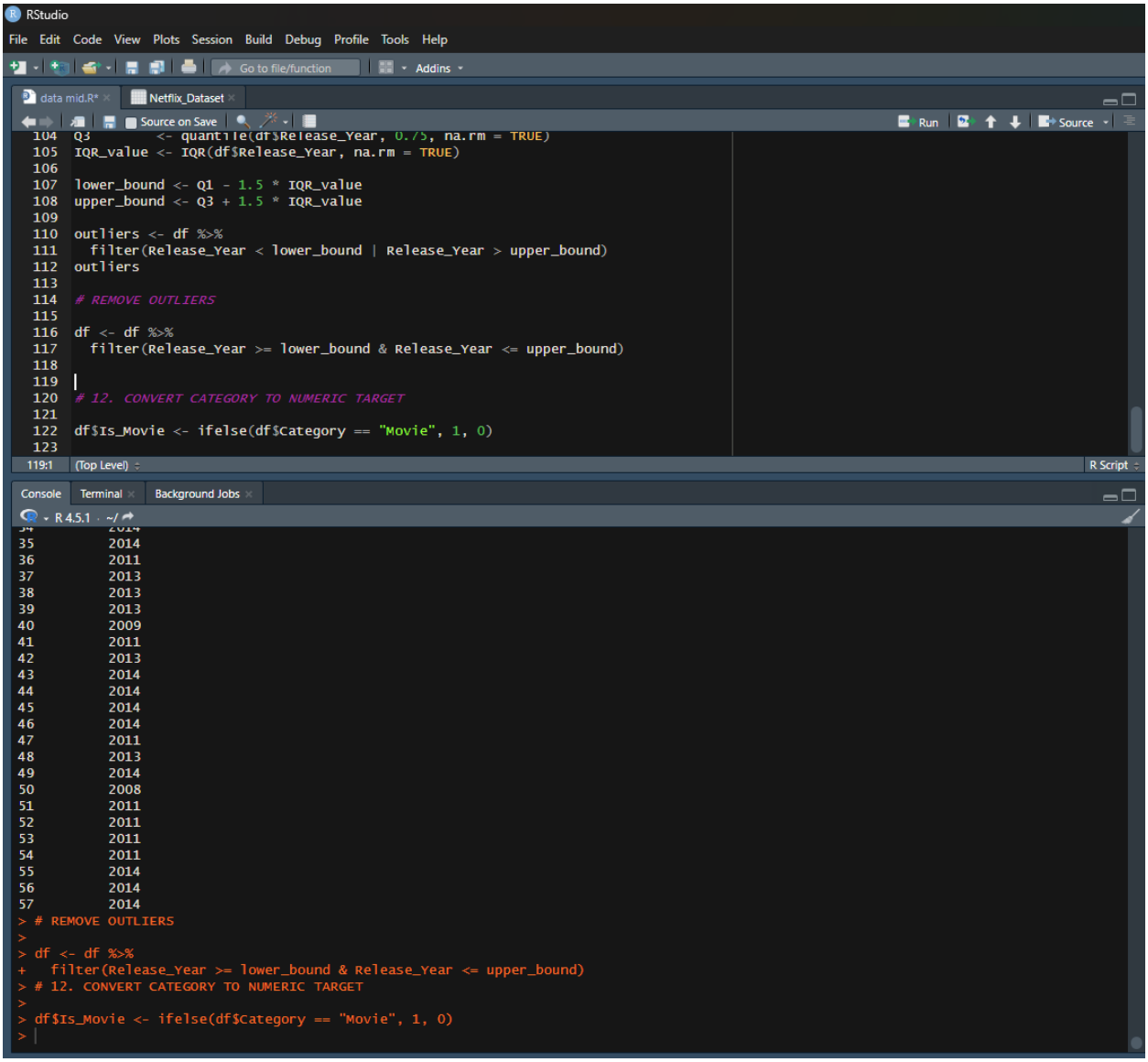
This code removes outliers by keeping only the rows where Release_Year falls within the acceptable range between the calculated lower and upper bounds.



13. Categorical to Numeric

```
df$Is_Movie <- ifelse(df$Category == "Movie", 1, 0)
```

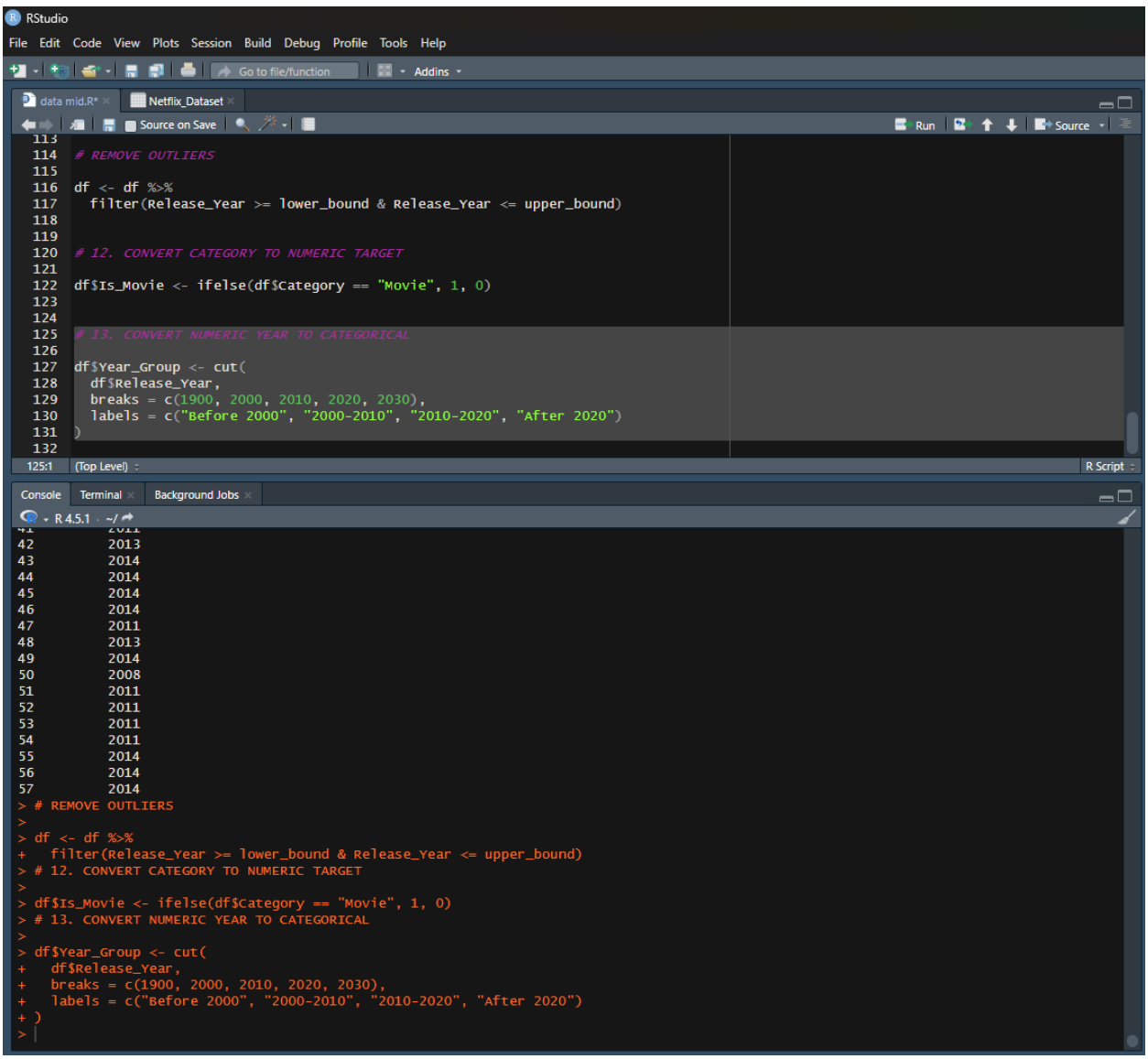
The Category column contains text labels (“Movie” or “TV Show”). To prepare for balancing and potential modeling, this field is converted into a numeric target variable (Is_Movie), where 1 represents a movie and 0 represents a TV show. This transformation ensures compatibility with numerical operations.



14. Numeric to Categorical

```
df$Year_Group <- cut(
  df$Release_Year,
  breaks = c(1900, 2000, 2010, 2020, 2030),
  labels = c("Before 2000", "2000-2010", "2010-2020", "After 2020")
)
```

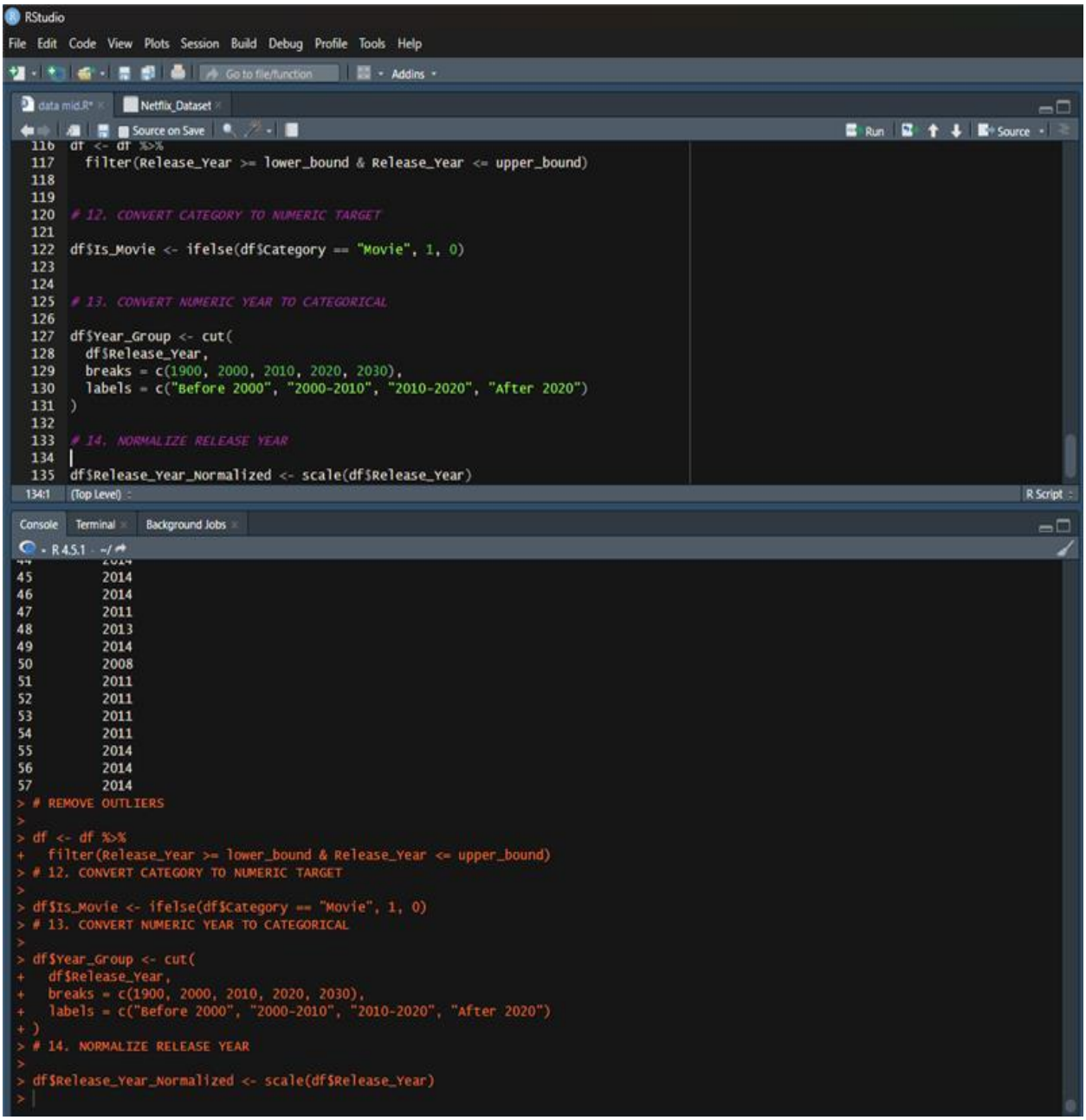
The Release_Year variable is grouped into meaningful time periods (Before 2000, 2000–2010, etc.) using the cut() function. Categorizing years enables easier comparison between time segments and supports more interpretable analysis.



15. Normalization

df$Release_Year_Normalized <- scale(df$Release_Year)

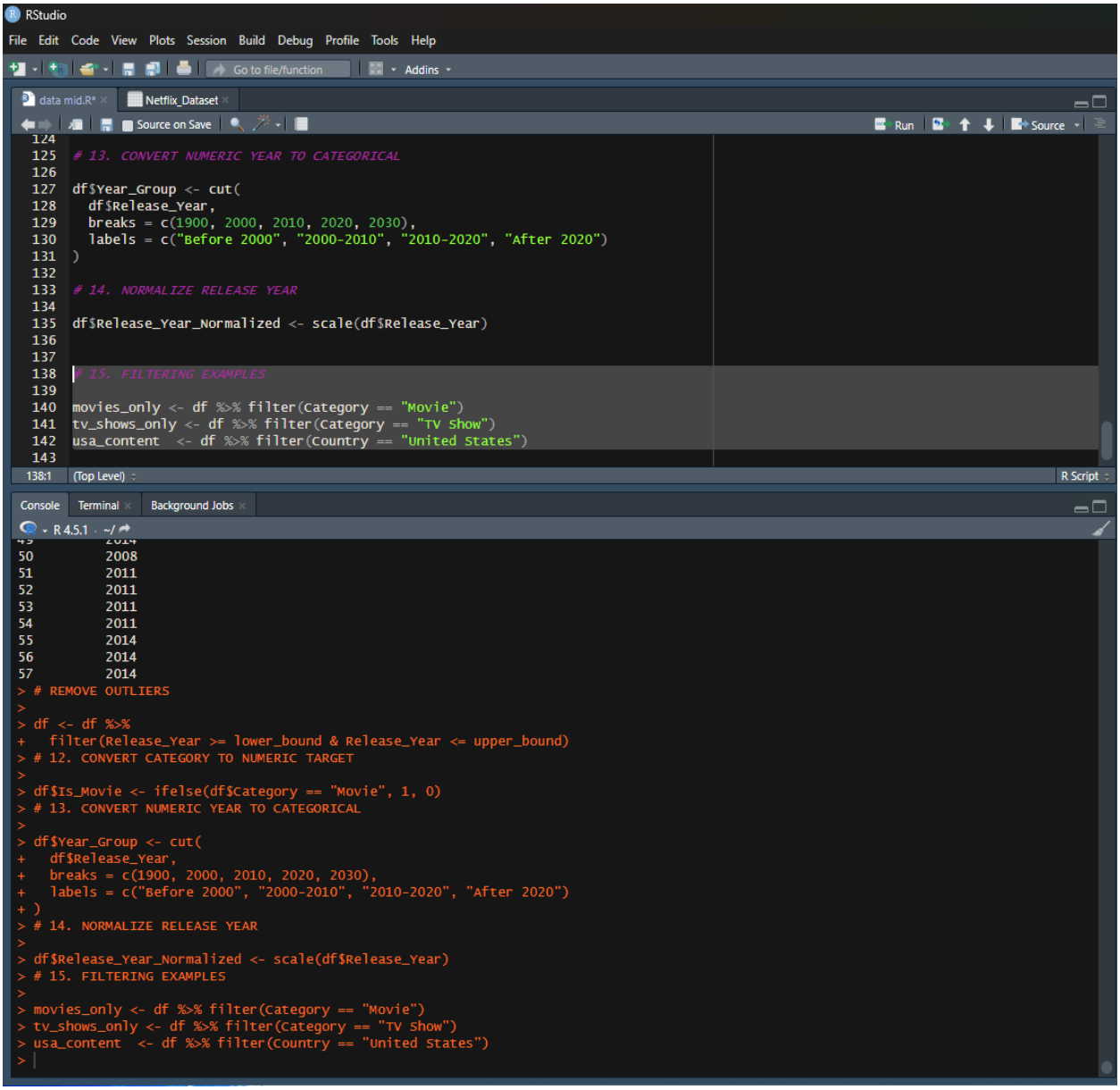
Normalization scales the Release_Year variable to have a mean of 0 and standard deviation of 1. This is useful when comparing numeric variables on equal footing or preparing data for certain algorithms that require standardized inputs.



16. Filtering

movies_only <- df %>% filter(Category == "Movie")
tv_shows_only <- df %>% filter(Category == "TV Show")
usa_content <- df %>% filter(Country == "United States")

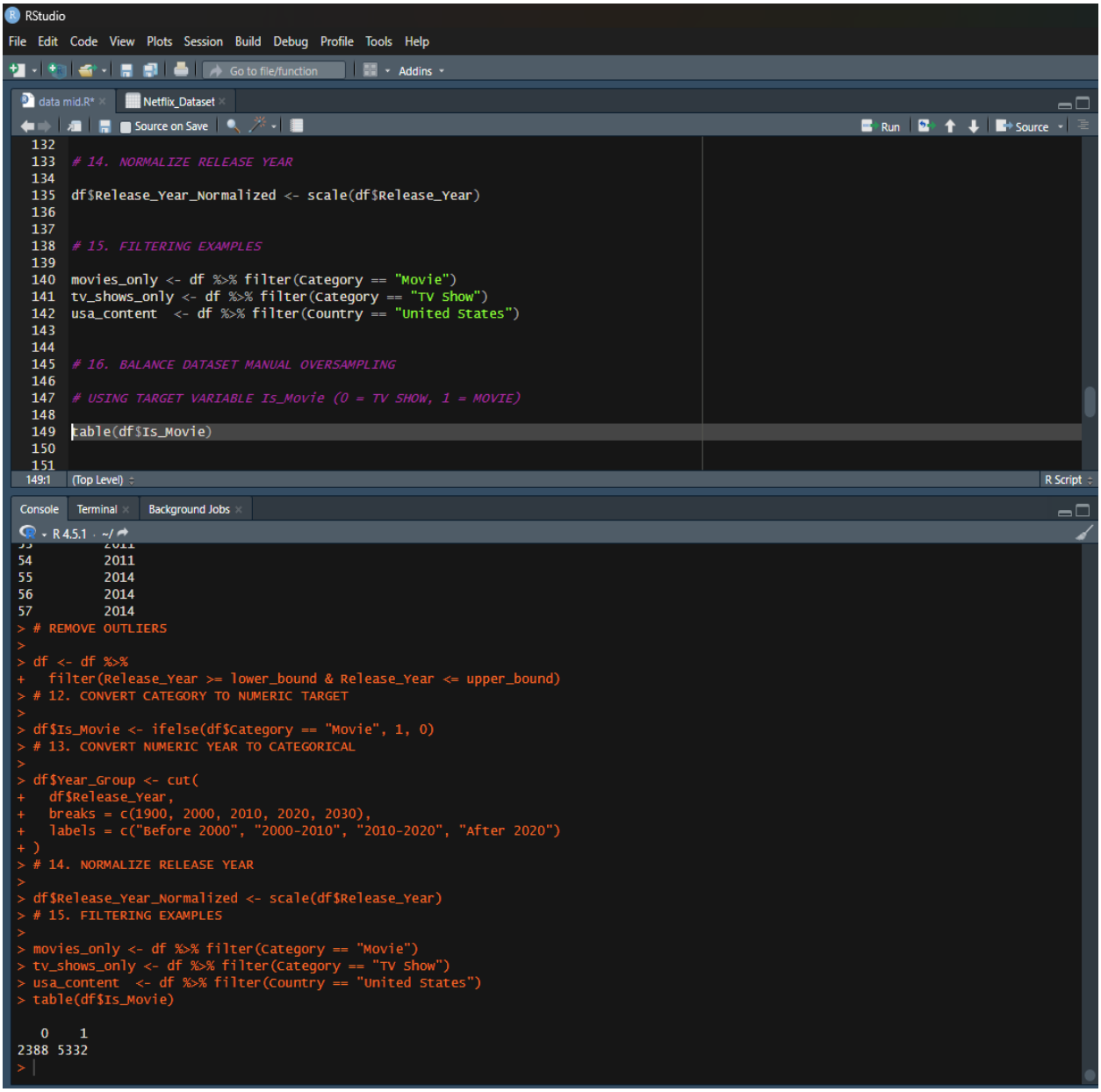
This section demonstrates how to filter the dataset based on specific conditions. Examples include extracting only movies, only TV shows, and titles produced in the United States. Filtering helps isolate subsets of interest for focused analysis.



Balancing the Dataset

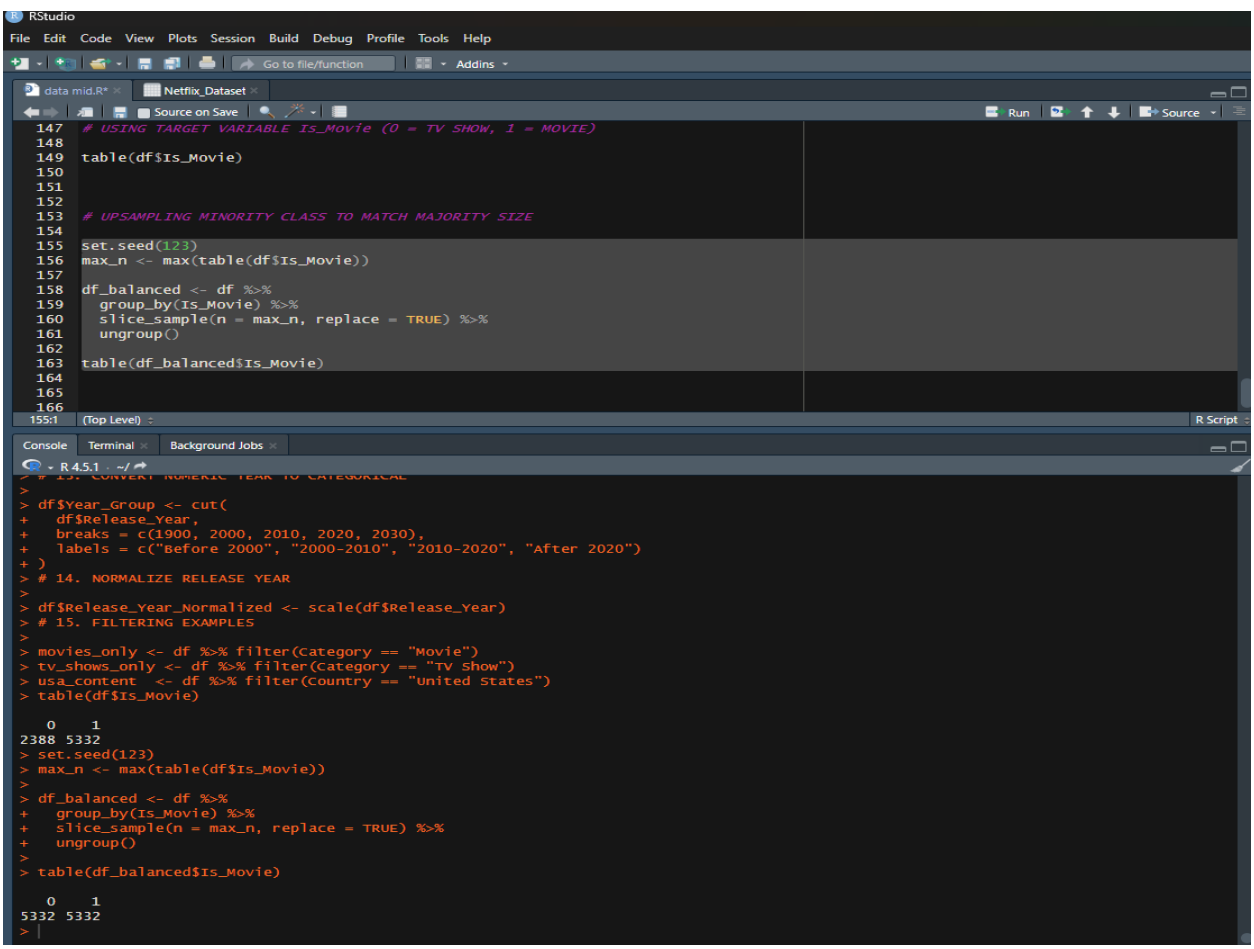
```
# USING TARGET VARIABLE Is_Movie (0 = TV SHOW, 1 = MOVIE)

table(df$Is_Movie)
```

The dataset is imbalanced, containing more movies than TV shows. Upsampling is used to duplicate rows of the minority class until both classes have equal representation. This ensures that subsequent analysis or modeling is not biased toward the majority class.



```
# UPSAMPLING MINORITY CLASS TO MATCH MAJORITY SIZE

set.seed(123)
max_n <- max(table(df$Is_Movie))

df_balanced <- df %>%
  group_by(Is_Movie) %>%
  slice_sample(n = max_n, replace = TRUE) %>%
  ungroup()

table(df_balanced$Is_Movie)
```

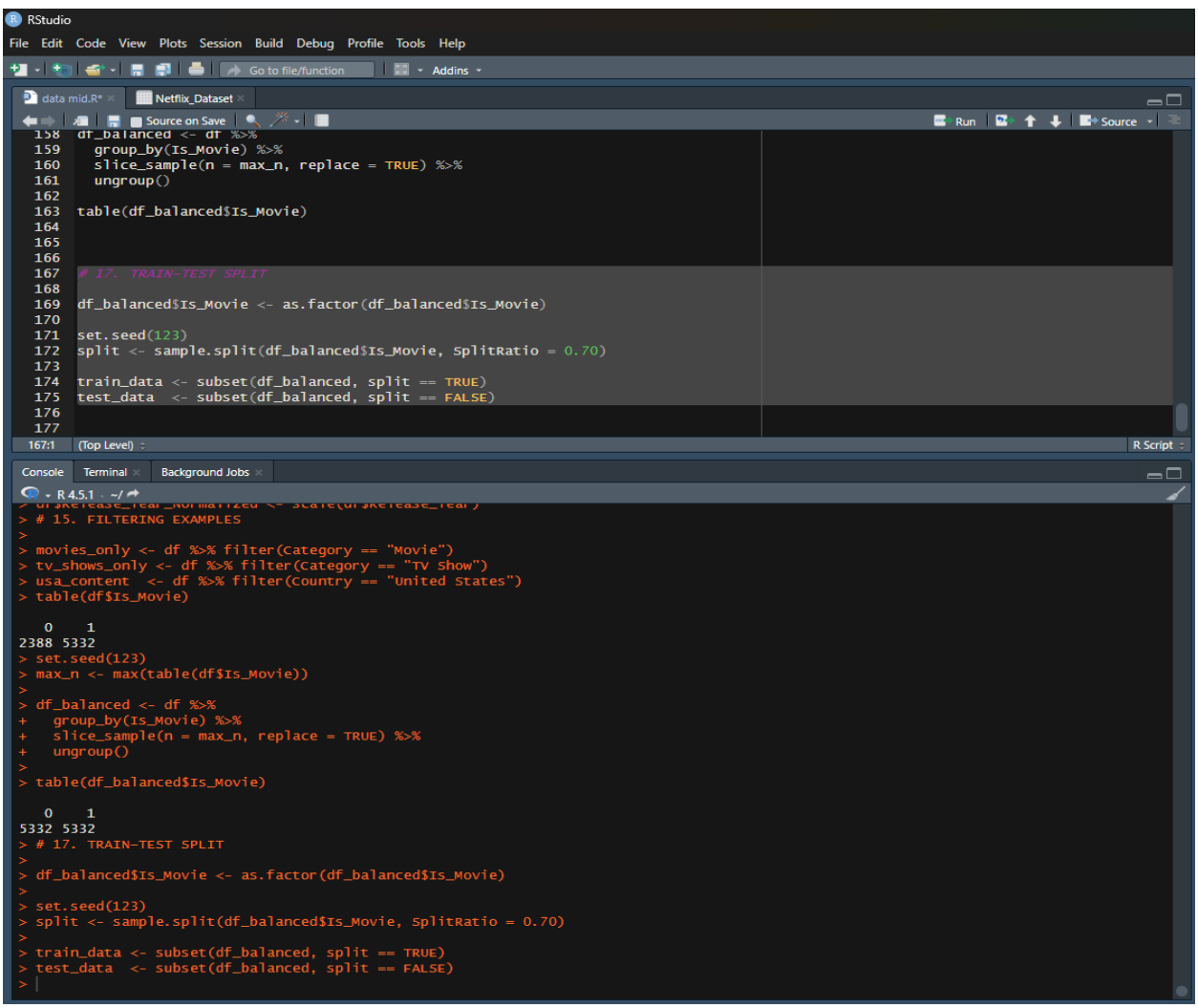
This code first sets a random seed to make the sampling results reproducible. It then finds the size of the largest class in the Is_Movie column. Using that number, the dataset is grouped by Is_Movie, and each group is randomly sampled with replacement so that every class ends up having the same number of rows as the largest class. After ungrouping, the data becomes a balanced dataset with equal representation from each class. Finally, the table() function shows the new class counts to confirm the balancing.



Train–Test Split

```
df_balanced$Is_Movie <- as.factor(df_balanced$Is_Movie)


set.seed(123)
split <- sample.split(df_balanced$Is_Movie, SplitRatio = 0.70)


train_data <- subset(df_balanced, split == TRUE)
test_data  <- subset(df_balanced, split == FALSE)
```

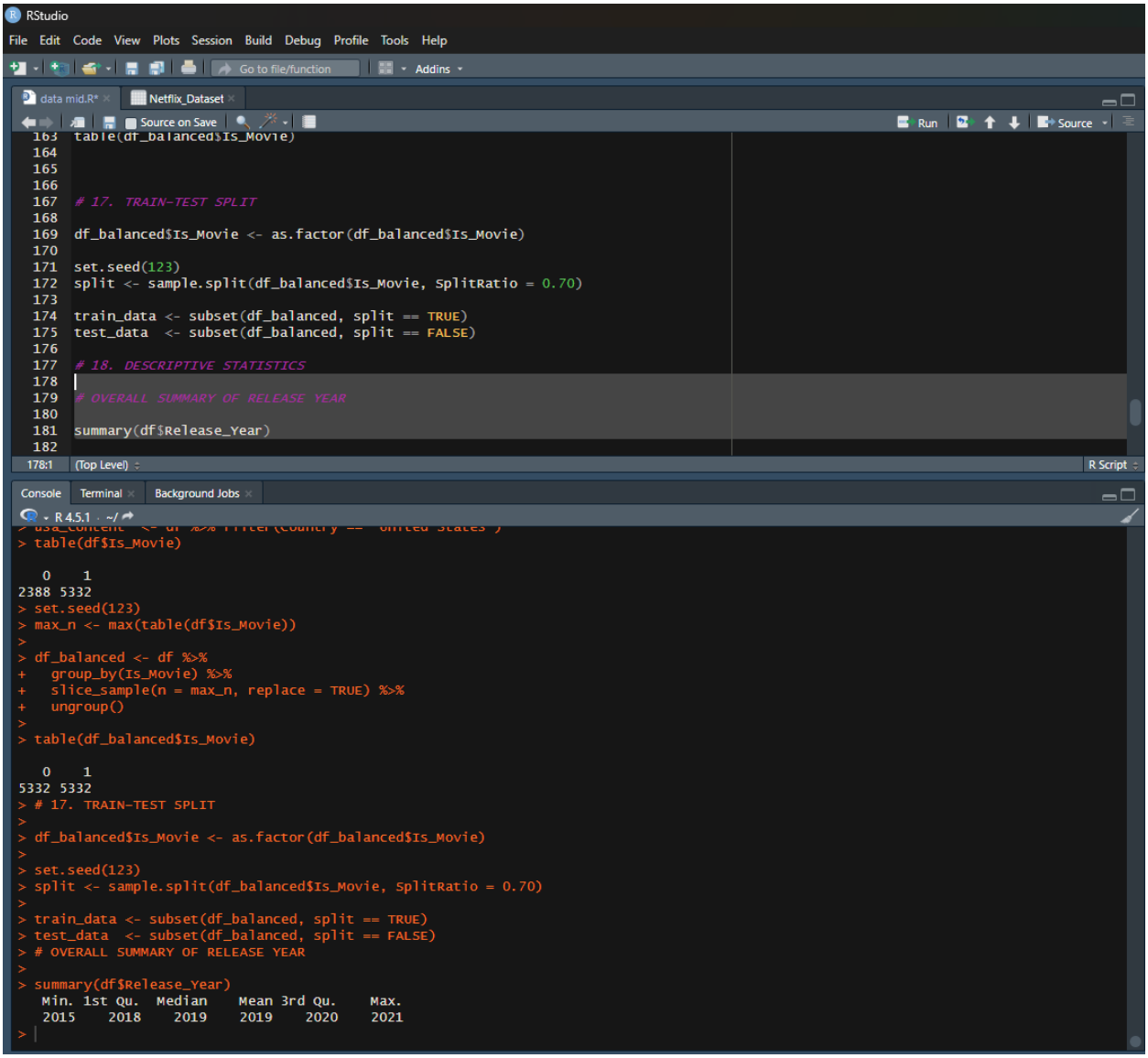
The balanced dataset is split into training and testing sets using a 70 - 30 ratio. Setting a random seed ensures reproducibility, meaning the same split is obtained each time the code is run. Train-test splitting is a standard practice when preparing data for machine learning.



Descriptive Statistics

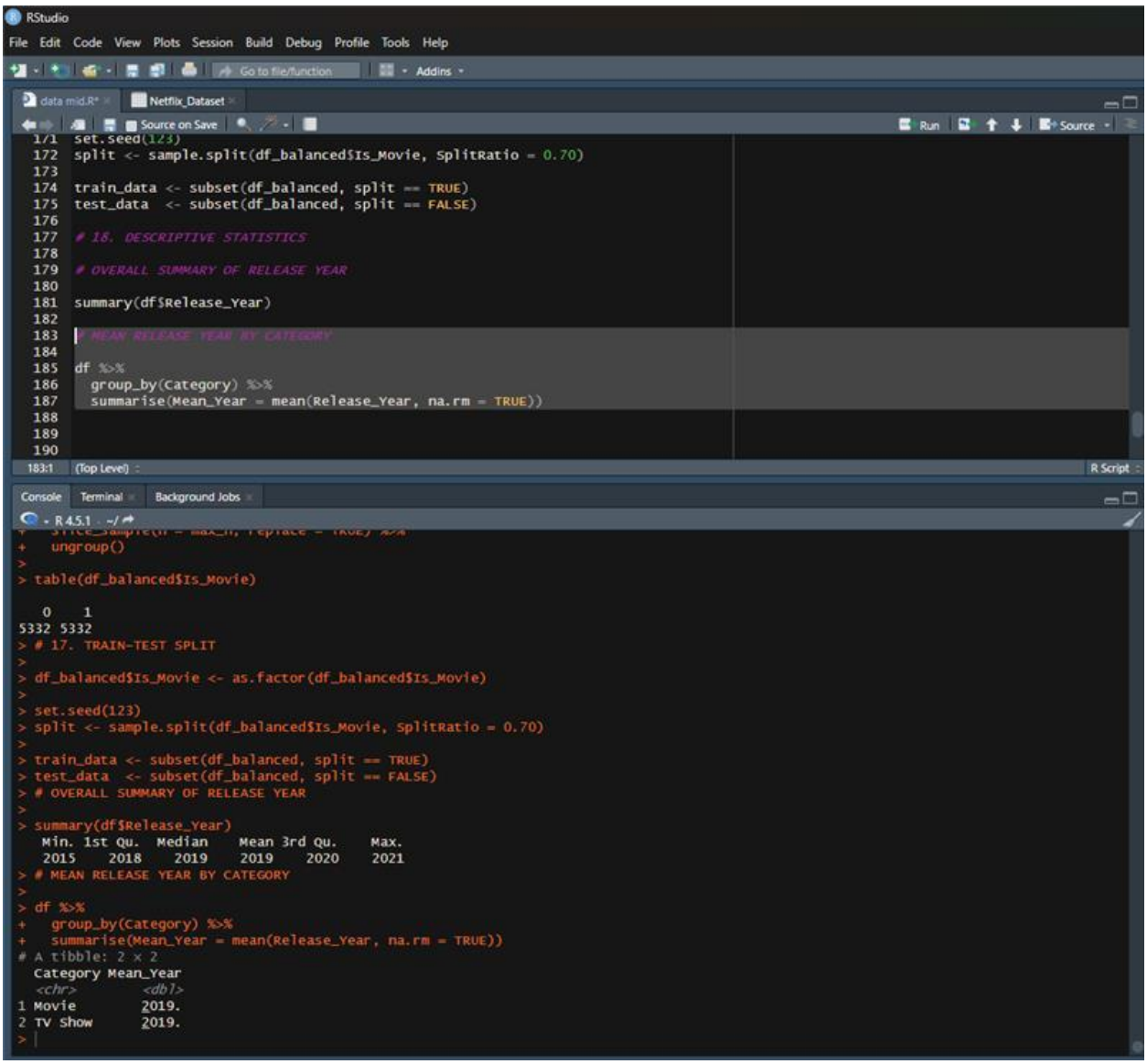
```
# OVERALL SUMMARY OF RELEASE YEAR

summary(df$Release_Year)
```

This block of code produces descriptive statistics about the release years in your dataset. It starts by using summary(df$Release_Year), which gives a quick overview of the minimum, maximum, median, and other key values for all release years.



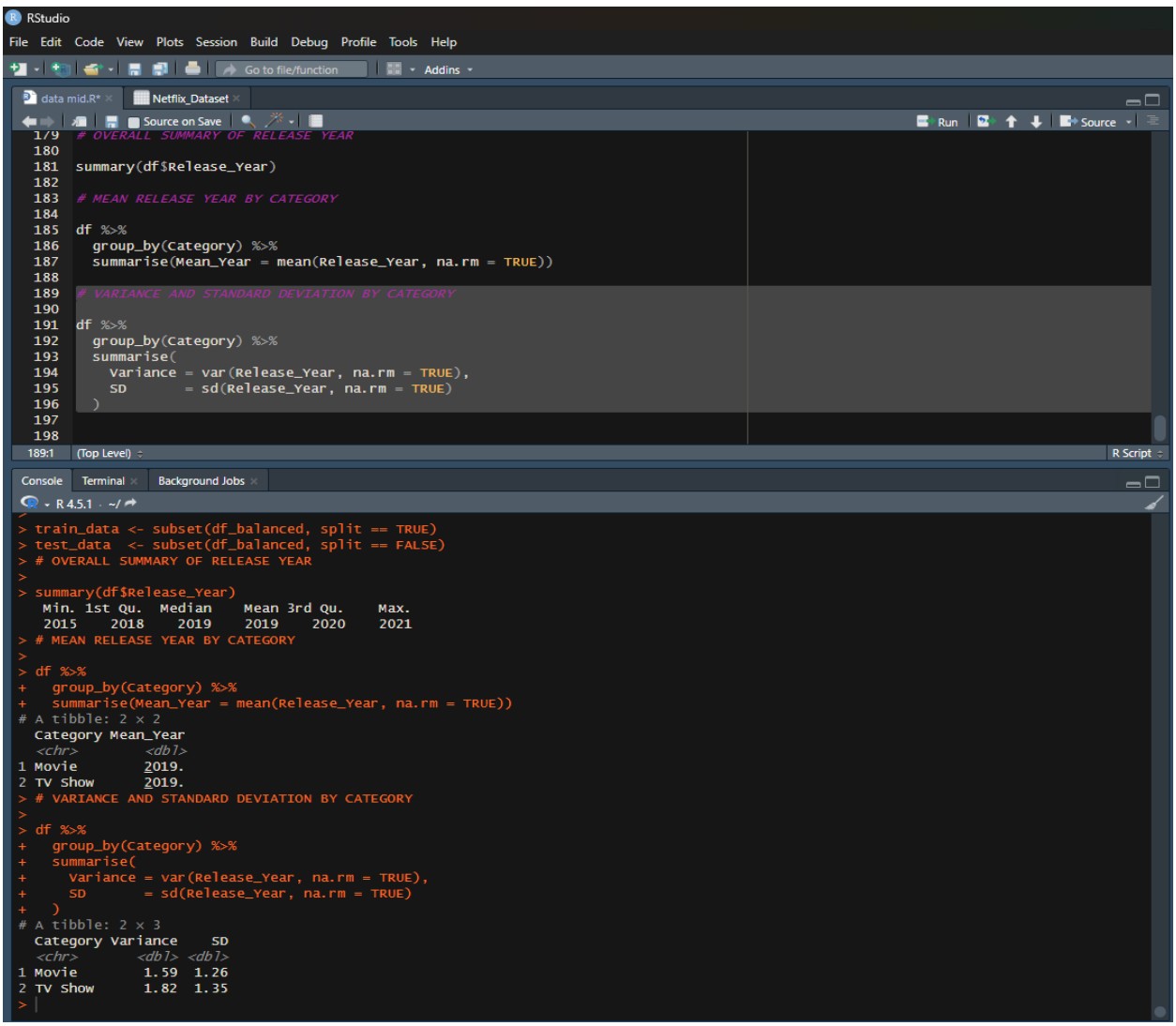
```
# MEAN RELEASE YEAR BY CATEGORY

df %>%
  group_by(Category) %>%
  summarise(Mean_Year = mean(Release_Year, na.rm = TRUE))
```

After that, the code calculates the average release year for each content category (such as Movies vs. TV Shows). It does this by grouping the data by Category and then taking the mean. The argument na.rm = TRUE is included to make sure missing values are removed before calculating otherwise, even a single NA would make the result invalid.



```
# VARIANCE AND STANDARD DEVIATION BY CATEGORY

df %>%
  group_by(Category) %>%
  summarise(
    Variance = var(Release_Year, na.rm = TRUE),
    SD       = sd(Release_Year, na.rm = TRUE)
  )
```

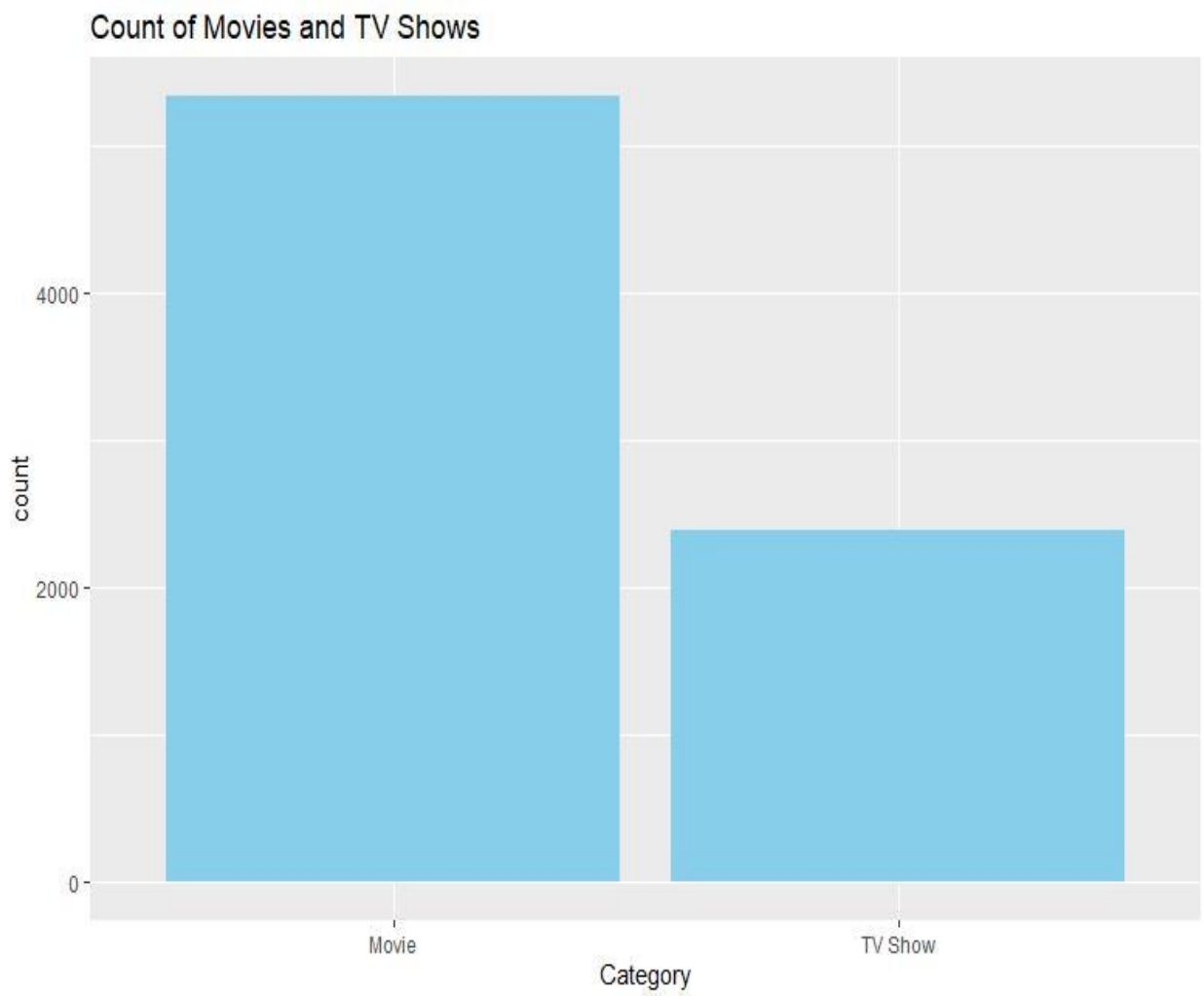
Finally, the code computes the variance and standard deviation of release years for each category. These measures show how spread out the release years are, again using na.rm = TRUE so that missing data doesn't interfere with the calculations. Overall, this section gives a clear picture of the distribution, central tendency, and variability of release years across different content types.



Exploratory Data Analysis

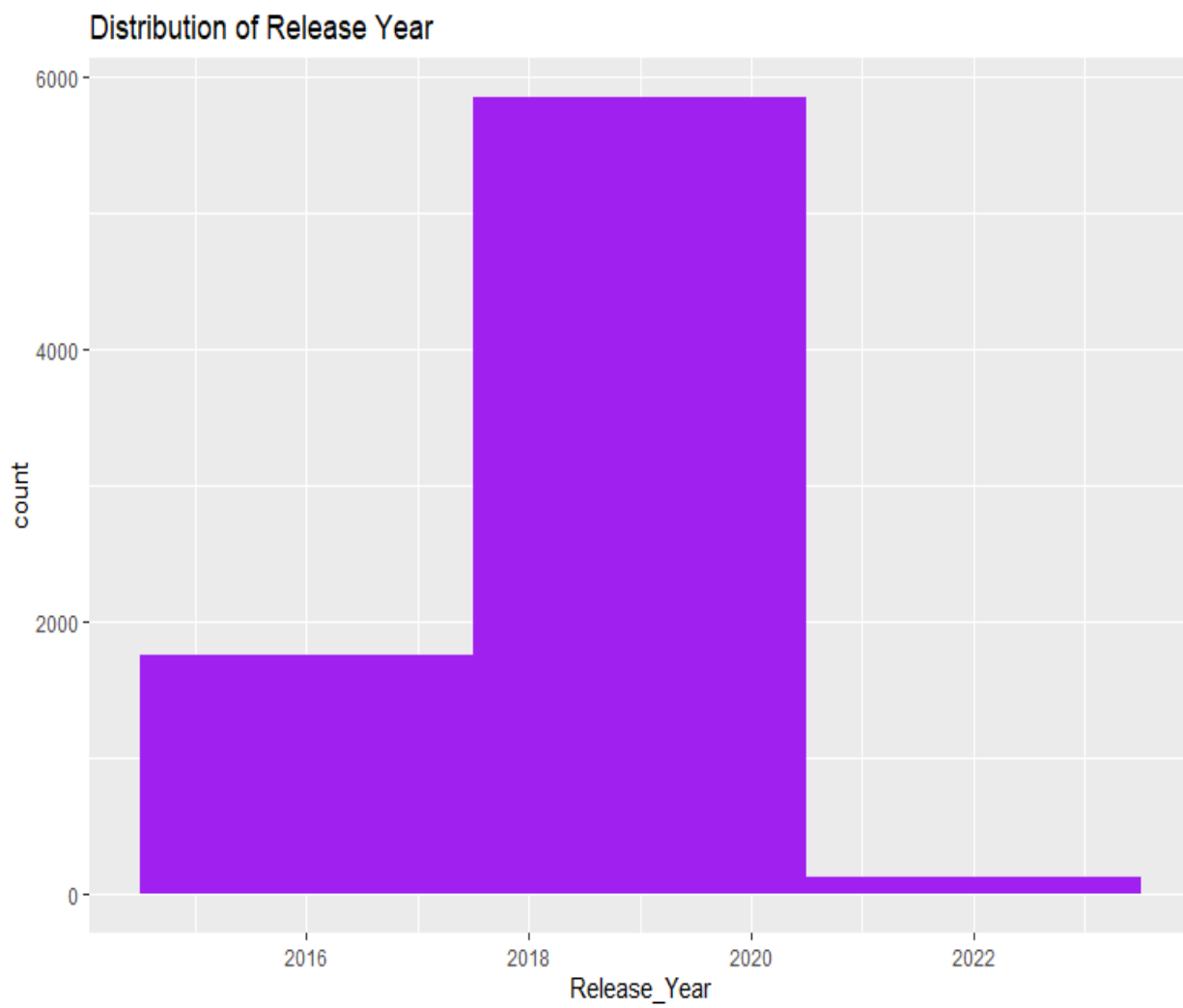
```
# CATEGORY DISTRIBUTION

ggplot(df, aes(Category)) +
  geom_bar(fill = "skyblue") +
  ggtitle("Count of Movies and TV Shows")
```

This part of the project creates simple visuals to help understand how the data is distributed. The first plot shows how many Movies and TV Shows are in the dataset using a bar chart (geom_bar()), where each bar represents the count of each category and is colored sky-blue.



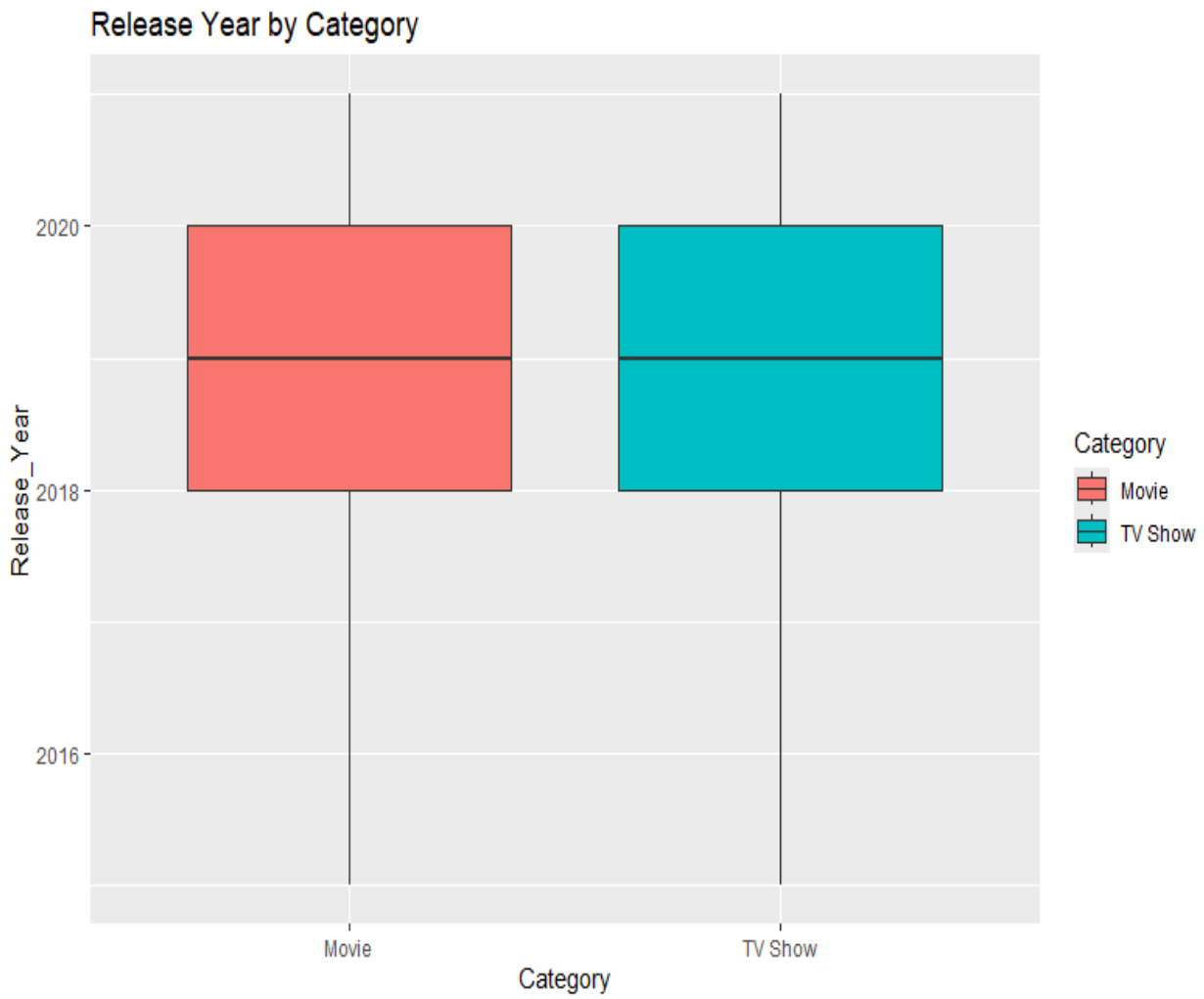
```
# HISTOGRAM OF RELEASE YEAR

ggplot(df, aes(Release_Year)) +
  geom_histogram(binwidth = 3, fill = "purple") +
  ggtitle("Distribution of Release Year")
```

The second plot is a histogram of the release years (geom_histogram(binwidth = 3)), which groups the years in 3-year intervals to show how content is spread across time; the bars are colored purple for clarity.



```
# BOXPLOT RELEASE YEAR BY CATEGORY

ggplot(df, aes(Category, Release_Year, fill = Category)) +
  geom_boxplot() +
  ggtitle("Release Year by Category")
```

The final plot is a boxplot (geom_boxplot()) comparing release years between Movies and TV Shows. The box shows the middle 50% of the data, the line inside the box represents the median, the whiskers show the range of most data, and any points outside are outliers. This allows you to quickly see which category tends to have older or newer content and how spread out the release years are. Overall, these three visuals provide an intuitive overview of category distribution and release-year patterns.



Conclusion

This project successfully carried out a complete and systematic preprocessing and exploratory analysis of the Netflix dataset using R, ensuring high data quality and analytical readiness. Through careful handling of missing values, removal of duplicates, correction of inconsistent formats, detection of invalid entries, and elimination of outliers, the dataset was transformed into a reliable and structured form. Feature engineering includes converting categorical variables to numeric, grouping release years, normalizing values, and filtering meaningful subsets further improved the dataset's usability. The class imbalance between movies and TV shows was effectively addressed through manual oversampling, followed by a clear separation into training and testing sets for future modeling tasks. Comprehensive descriptive statistics and visual exploratory analysis provided valuable insights into Netflix's content trends, distribution patterns, and temporal characteristics. Overall, the project delivered a clean, well-organized, and insight-rich dataset that not only enhances understanding of Netflix's global content library but also establishes a strong foundation for advanced predictive modeling, decision-making, and further data science applications.

Reference:

1. Dataset Link: Netflix Data